

LABORATORY 06 - APPLICATION LAYER PROTOCOLS, BASE PLATFORM AND NETWORK LAYER

Laboratory 6 corresponding to the third term

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NETWORK ARCHITECTURE AND SERVICES

GROUP 3

Bogotá DC, Colombia

2025 - 2

Lab No.06 - APPLICATION LAYER PROTOCOLS, BASE PLATFORM AND NETWORK LAYER

Objectives

Install and configure basic software for web servers.

Tools to be used

- Computers
- Internet access

Introduction

In today's enterprises, IT infrastructure is a fundamental component that enables the operation of multiple business services. Organizations rely on interconnected systems to support internal and external communication, web applications, data processing, and storage. This infrastructure typically includes user workstations, physical and virtual servers, and both wired and wireless networks that ensure seamless service availability.

To achieve proper communication across all systems, networking devices such as Layer 2 switches, Layer 3 switches, routers, and wireless access points must be configured correctly. These devices not only interconnect users and servers but also ensure efficient routing between different network segments, optimizing traffic flow and enabling secure access to the Internet. Configuring routing, addressing, and connectivity verification is therefore crucial in a real-world environment.

Web services play a vital role within the application layer of modern networks. They allow users to reach websites, academic platforms, business portals, and internal applications essential for organizational operations. These services may be deployed locally in the company's data center or hosted in the cloud depending on performance, scalability, and budget requirements. A well-implemented web infrastructure increases system reliability, flexibility, and user experience.

In this laboratory, students will simulate real operational scenarios by deploying and configuring web server software across different operating systems, including Solaris, Slackware, and Windows Server. Additionally, routing configurations and DNS services will be implemented to ensure global connectivity and access to websites through domain names.

By documenting and testing these configurations, students gain practical knowledge directly aligned with the needs of IT administration and network management in professional environments.

Theoretical Framework

The **Application Layer**, found at the top of the TCP/IP model, is responsible for providing network services directly to end users. Protocols such as **HTTP** and **HTTPS** enable the transfer of web pages between clients and servers, while **DNS** resolves domain names into IP addresses, facilitating user-friendly communication. Other important protocols like **FTP**, **SMTP**, and **IMAP** support file transfers and email services. Understanding this layer is essential when configuring web hosting solutions, as it ensures that user requests are correctly processed and delivered.

The **Network Layer** (Layer 3 of the OSI model) provides logical addressing and routing between remote networks using protocols such as **IP**, **ICMP**, and **ARP**. Routers use static or dynamic routing techniques to determine the most efficient path for data transmission. In practical terms, without correct routing configuration, devices located in different subnets cannot communicate, even if physically connected.

Web hosting relies on dedicated daemons such as **Apache HTTP Server**, **Nginx**, and **IIS**. These servers listen for incoming client requests and deliver static or dynamic content back to users through the HTTP protocol. For dynamic web applications, languages such as **PHP** interact with database management systems like **PostgreSQL**, allowing storage, processing, and retrieval of student records, business information, and more. Proper configuration ensures secure database connectivity and reliable backend operation.

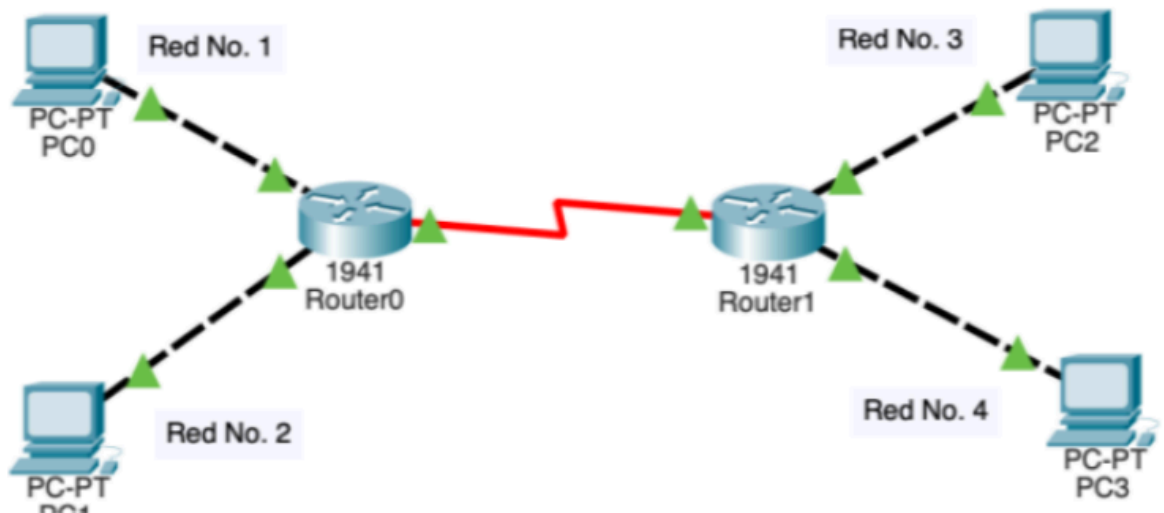
Finally, **DNS services** make navigation intuitive by mapping domain names to their corresponding IP addresses. Depending on organizational needs, DNS may manage multiple domains and subdomains, enabling services such as virtual hosting. This allows several websites to reside on a single web server while being accessed through different domain names. DNS reliability is essential to guarantee accessibility and maintain functional and scalable IT environments.

Simulations

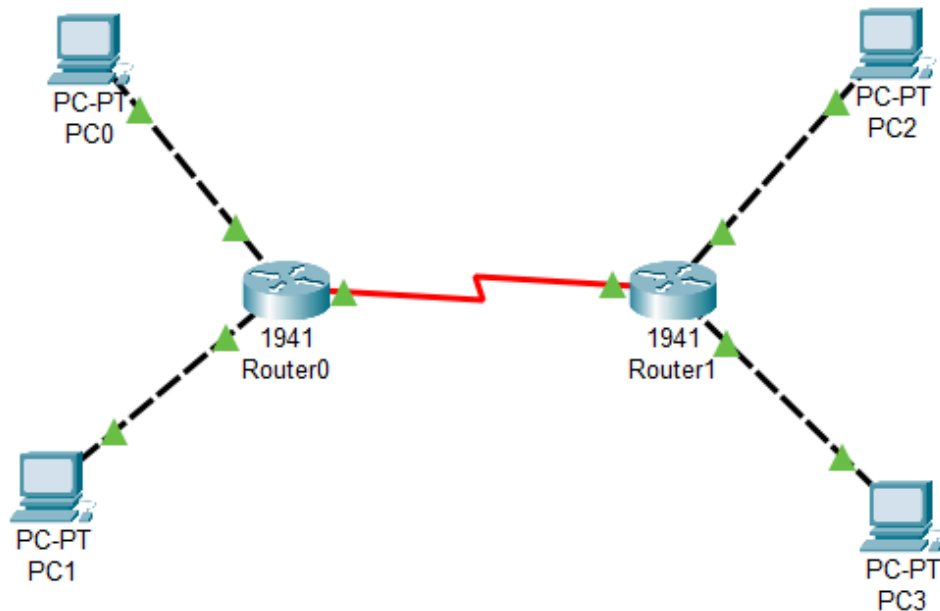
Conduct the following tests individually or in groups, as instructed by the professor, and document your experience.

1. Router Simulation

Open Packet Tracer and create a network following the diagram.



Santiago's diagram:



Now, to configure it, connect from a console similarly to how you do with switches. (The blue cable is a console cable and connects to the serial port – RS232 of each device).



Configure the networks to allow addressing the specified hosts.

Networks:

For the serial interconnection network, use 20.0.0.0/8.

– Student 01(Santiago):

* 132.18.0.0/16

* 132.19.0.0/16

* 132.20.0.0/16

* 132.21.0.0/16

Access each router from the computer by going to the desktop tab and selecting the terminal option. Do not modify the default parameters (this is similar to what we will do going forward with the physical routers using the light blue cable at each workstation in the lab). Manually configure the router, so when prompted with "Continue with configuration dialog? [yes/no]:", select n, then enter privileged mode and proceed to configuration mode.

Student 01 (Santiago)

On Router0, editing from PC using the console cable, I configured the IPs of the 3 interfaces that are being used

```
Router>
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router0(config)#hostname Router0
Router0(config)#interface gig0/0
Router0(config-if)#ip address 132.18.0.1 255.255.0.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#interface gig0/1
Router0(config-if)#ip address 132.19.0.1 255.255.0.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#interface serial0/0/0
Router0(config-if)#ip address 20.0.0.1 255.0.0.0
Router0(config-if)#clock rate 64000
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#
```

I did the same with the interfaces of Router1

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Router1
Router1(config)#interface gig0/0
Router1(config-if)#ip address 132.20.0.1 255.255.0.0
Router1(config-if)#no shutdown
Router1(config-if)#exit
Router1(config)#interface gig0/1
Router1(config-if)#ip address 132.21.0.1 255.255.0.0
Router1(config-if)#no shutdown
Router1(config-if)#exit
Router1(config)#interface serial0/1/0
Router1(config-if)#ip address 20.0.0.2 255.0.0.0
Router1(config-if)#no shutdown
Router1(config-if)#exit
Router1(config)#end
Router1#
%SYS-5-CONFIG_I: Configured from console by console
Router1#
```

Now configure:

- Name of Router0: Your name.
- Name of Router1: Your surname.
- Message of day: "For exclusive use by RECO students".
- Screen synchronization.
- Block external server command lookup.
- Interface descriptions.
- Device passwords:
 - * Privilege mode: Clave E
 - * Consult: Clave C
 - * Remote terminal: Clave T

Now I modified the hostnames of the routers, changing router0 to santiago and router1 to amaya

```
Router0#configure t
Enter configuration commands, one per line. End with CNTL/Z.
Router0(config)#hostname santiago
santiago(config)#
```

```
Router1#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router1(config)#hostname amaya
amaya(config)#
```

I posted the message of the day on both routers

```
santiago>enable
santiago#config t
Enter configuration commands, one per line. End with CNTL/Z.
santiago(config)#banner motd "For exclusive use by RECO students."
santiago(config)#
```

```

amaya>enable
amaya#config t
Enter configuration commands, one per line. End with CNTL/Z.
amaya(config)#banner motd "For exclusive use by RECO students."
amaya(config)#

```

To achieve screen synchronization, we enter the console line configuration, then enable synchronization and exit to return to global configuration mode.

```

santiago(config)#line console 0
santiago(config-line)#logging synchronous
santiago(config-line)#exit
santiago(config)#

```

```

amaya(config)#line console 0
amaya(config-line)#logging synchronous
amaya(config-line)#exit
amaya(config)#

```

Now you should block the lookup of commands on the external server, that is, the automatic DNS lookup is blocked.

```

santiago(config)#no ip domain-lookup
santiago(config)#

```

```

amaya(config)#no ip domain-lookup
amaya(config)#

```

Now we must add a description to each interface to document its use.

```

santiago(config)#interface gigabitethernet0/0
santiago(config-if)#description connection to PC1
santiago(config-if)#exit
santiago(config)#interface gigabitethernet0/1
santiago(config-if)#description connection to PC0
santiago(config-if)#exit
santiago(config)#interface serial0/0/0
santiago(config-if)#description connection to Router(amaya)
santiago(config-if)#exit
santiago(config)#

```

```

amaya(config)#interface gigabitethernet0/0
amaya(config-if)#description connect to PC3
amaya(config-if)#exit
amaya(config)#interface gigabitethernet0/1
amaya(config-if)#description connect to PC2
amaya(config-if)#exit
amaya(config)#interface serial0/1/0
amaya(config-if)#description connect to Router(santiago)
amaya(config-if)#exit
amaya(config)#

```

Now we proceed to configure the passwords for the different access levels, for the privileged mode we use secret, since it allows strong encryption and for the query and remote terminal modes we use the basic command

```

santiago(config)#enable secret Clave_E
santiago(config)#

```

```

amaya(config)#enable secret Clave_E
amaya(config)#

```

```
santiago(config)#line console 0
santiago(config-line)#password Clave_C
santiago(config-line)#login
santiago(config-line)#exit
santiago(config)#
```

```
amaya(config)#line console 0
amaya(config-line)#password Clave_C
amaya(config-line)#login
amaya(config-line)#exit
amaya(config)#
```

```
santiago(config)#line vty 0 4
santiago(config-line)#password Clave_T
santiago(config-line)#login
santiago(config-line)#exit
santiago(config)#
```

```
amaya(config)#line vty 0 4
amaya(config-line)#password Clave_T
amaya(config-line)#login
amaya(config-line)#exit
amaya(config)#
```

Finally we save the configuration so that it is maintained after a reboot.

```
santiago#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
santiago#
```

```
amaya#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
amaya#
```

Test connectivity between the PCs on the same LAN and to different networks. Which ones work and which ones don't?

First we must configure the IP, mask and gateway for each PC

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

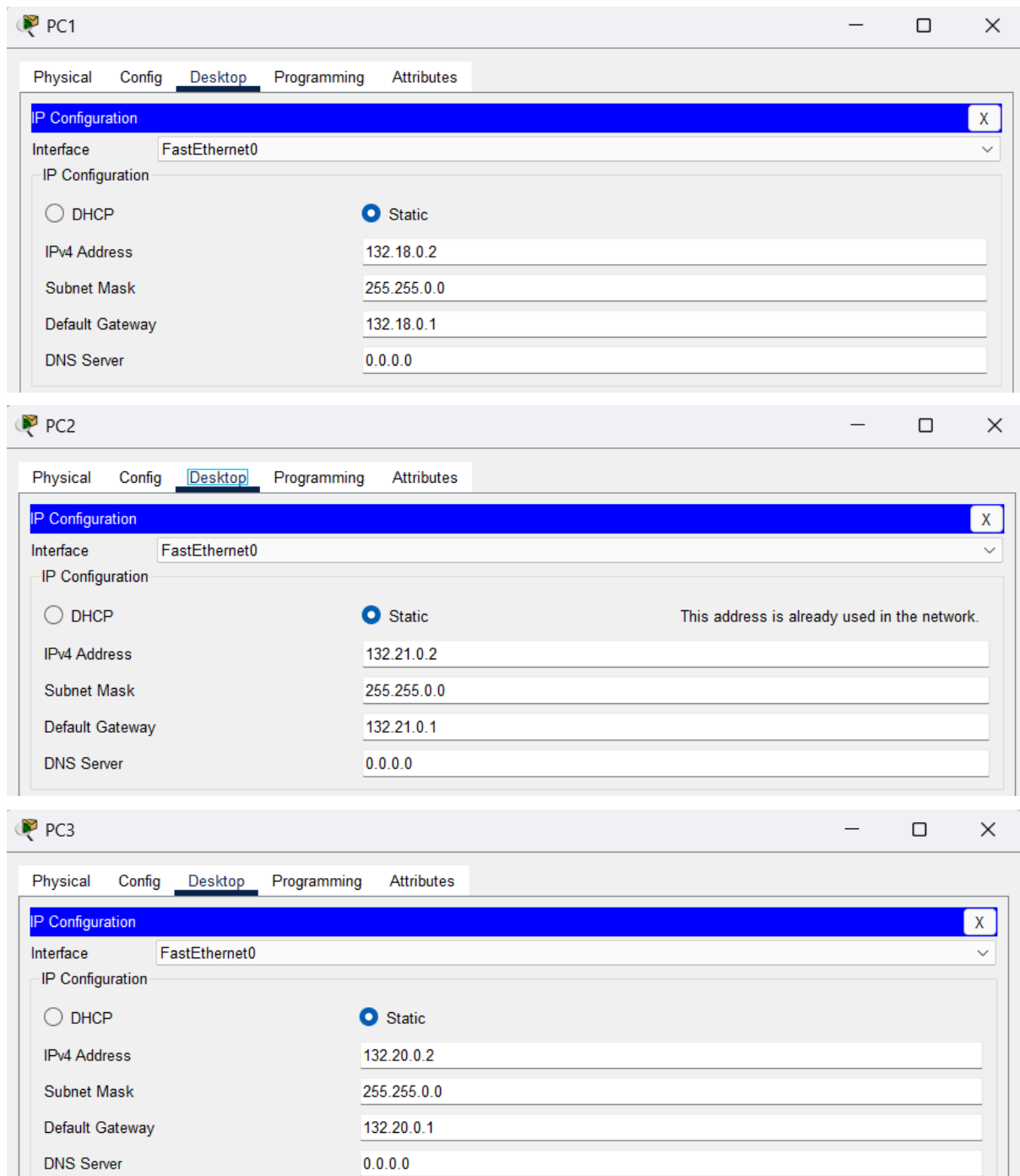
☐ DHCP ☒ Static

IPv4 Address 132.19.0.2

Subnet Mask 255.255.0.0

Default Gateway 132.19.0.1

DNS Server 0.0.0.0



Now we test connectivity: Ping on the same LAN

```
C:\>ping 132.18.0.2

Pinging 132.18.0.2 with 32 bytes of data:

Request timed out.
Reply from 132.18.0.2: bytes=32 time<1ms TTL=127
Reply from 132.18.0.2: bytes=32 time<1ms TTL=127
Reply from 132.18.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 132.18.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

ping on another LAN

```
C:\>ping 132.21.0.2

Pinging 132.21.0.2 with 32 bytes of data:

Reply from 132.19.0.1: Destination host unreachable.
Reply from 132.19.0.1: Destination host unreachable.
Reply from 132.19.0.1: Destination host unreachable.
Reply from 132.19.0.1: Destination host unreachable.

Ping statistics for 132.21.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

The connection fails when trying to ping a PC located on another network.

Configure the routers with static routes so that all devices on the network can ping each other. Note: On each router, include the routes (exit interface) needed to reach networks that are not directly connected.

Every router must have static routes to reach networks not directly connected to it. If we start with the santiago router, both routes from the amaya router are reached via the Serial 0/0/0 interface.

```
santiago#config t
Enter configuration commands, one per line. End with CNTL/Z.
santiago(config)#ip route 132.20.0.0 255.255.0.0 serial 0/0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
santiago(config)#ip route 132.21.0.0 255.255.0.0 serial 0/0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
santiago(config)#end
santiago#
%SYS-5-CONFIG_I: Configured from console by console

santiago#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
santiago#|
```

And from the amaya router using the serial 0/1/0 interface

```
amaya#config t
Enter configuration commands, one per line. End with CNTL/Z.
amaya(config)#ip route 132.18.0.0 255.255.0.0 serial 0/1/0
%Default route without gateway, if not a point-to-point interface, may impact performance
amaya(config)#ip route 132.19.0.0 255.255.0.0 serial 0/1/0
%Default route without gateway, if not a point-to-point interface, may impact performance
amaya(config)#end
amaya#
%SYS-5-CONFIG_I: Configured from console by console

amaya#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
amaya#|
```

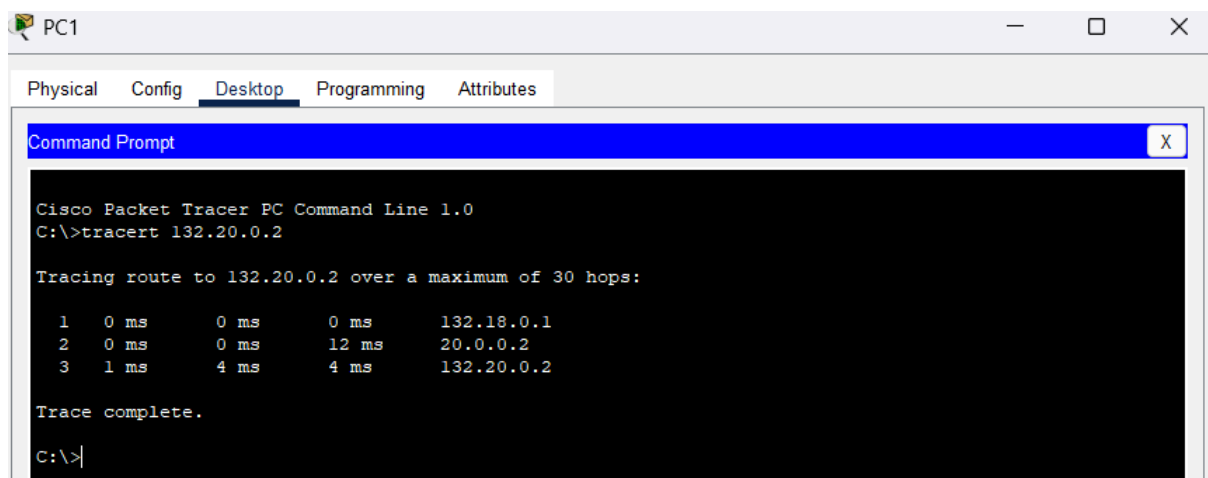
We verify that the routes have been added to the routing table

```
santiago#show ip route static
S    132.20.0.0/16 is directly connected, Serial0/0/0
S    132.21.0.0/16 is directly connected, Serial0/0/0
santiago#
```

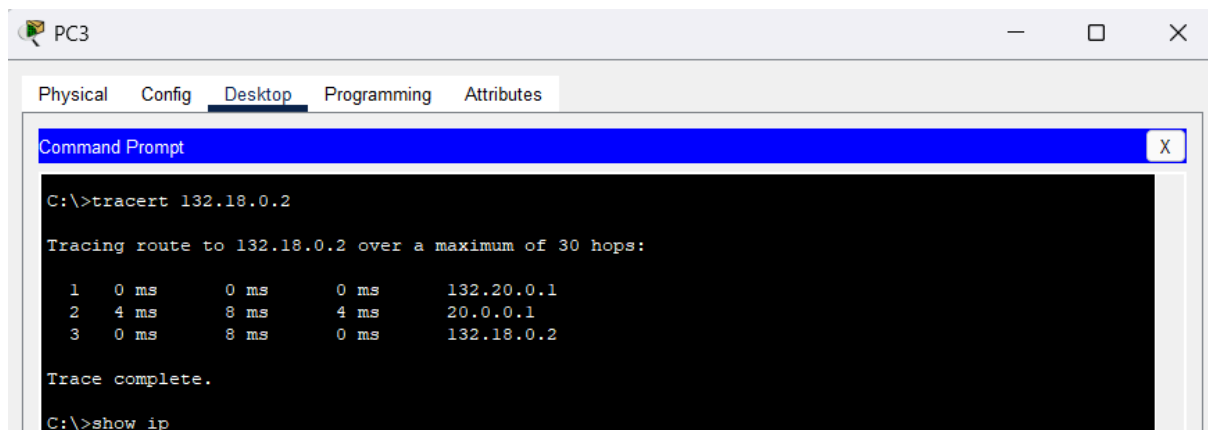
```
amaya#show ip route static
S    132.18.0.0/16 is directly connected, Serial0/1/0
S    132.19.0.0/16 is directly connected, Serial0/1/0
amaya#
```

Using the ICMP traceroute command, try to verify communication between the two computers.

From PC1 to PC3

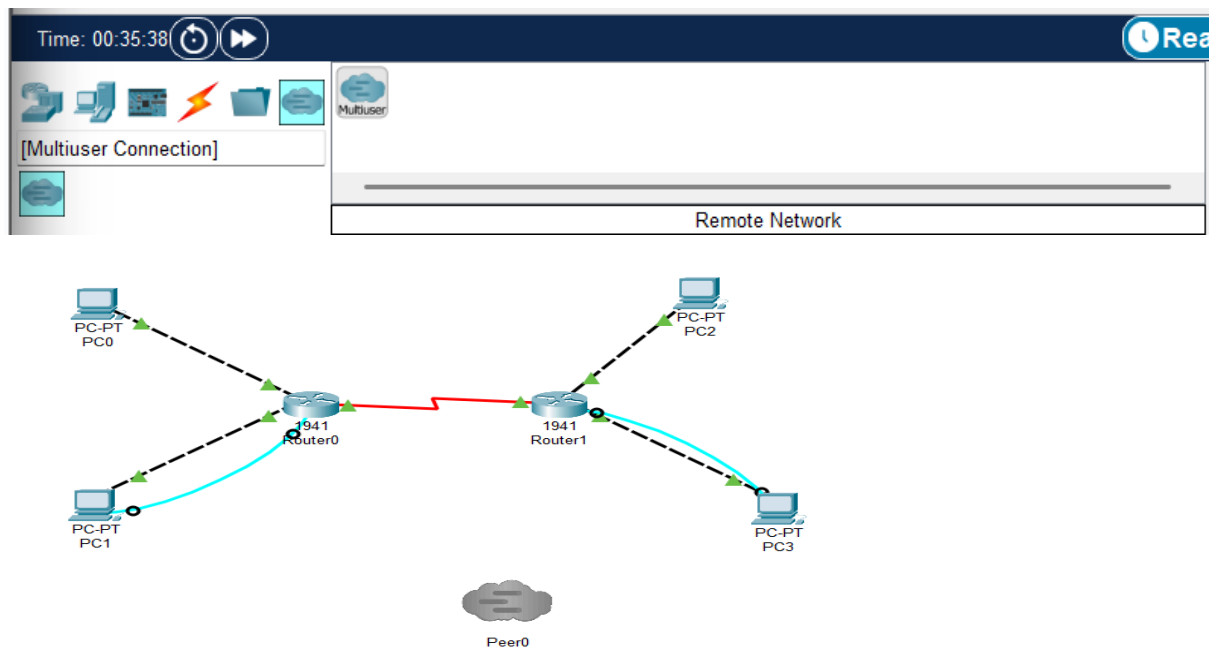


From PC3 to PC1



Connect the setups using the multiuser option so that all computers on the network are visible, and verify the path that the packets follow. Connect the setups through Router1.

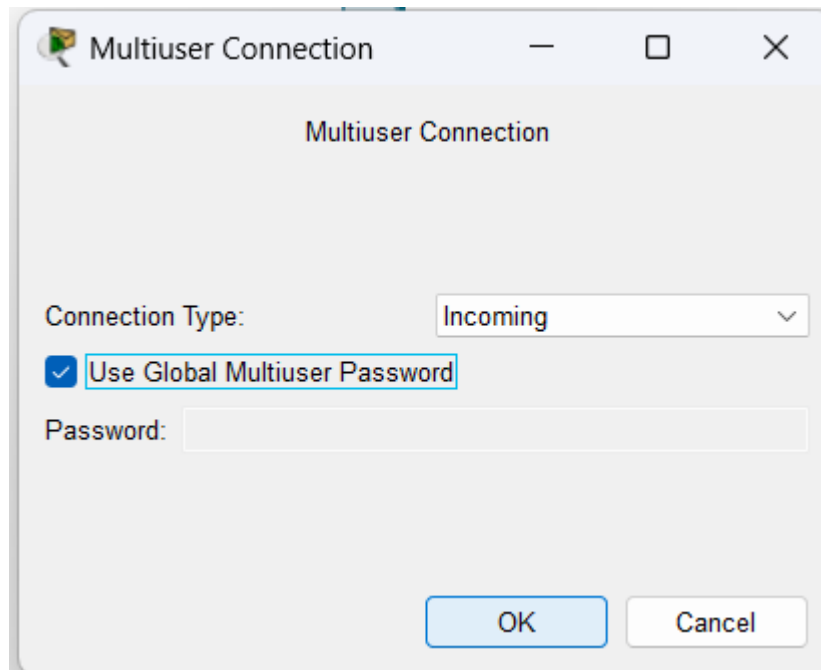
First, you must add the multi-user connection using Multiuser connection



In extensions, we click on multiuser and then on listen...

The screenshot shows the 'Multiuser Listen' dialog box. The title bar reads 'Multiuser Listen'. The 'Local Listening Address:' label is positioned to the left of a text area containing the following IP addresses and ports: 192.168.56.1:38000, 192.168.231.1:38000, 192.168.187.1:38000, 192.168.1.10:38000, and 127.0.0.1:38000. Below this, the 'Port Number' is set to 38000, and the 'Password' field is masked with dots. At the bottom, there are two sections: 'Existing Remote Networks' and 'New Remote Networks'. Each section contains three radio button options: 'Always Accept', 'Always Deny', and 'Prompt'. In both sections, the 'Prompt' option is selected. At the very bottom, there are three buttons: 'Stop Listening', 'OK', and 'Cancel'.

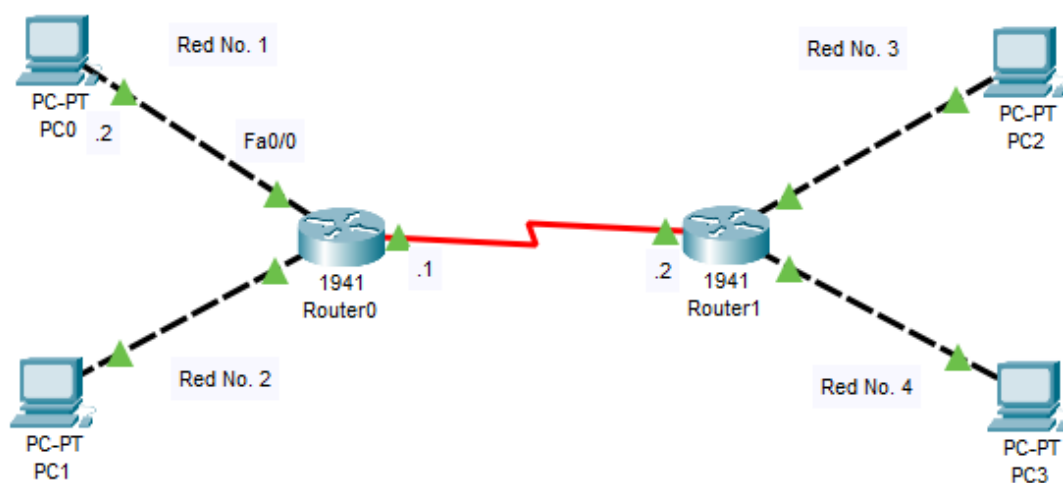
We click on the cloud and select the connection type; if it's a server, it should be left as incoming.



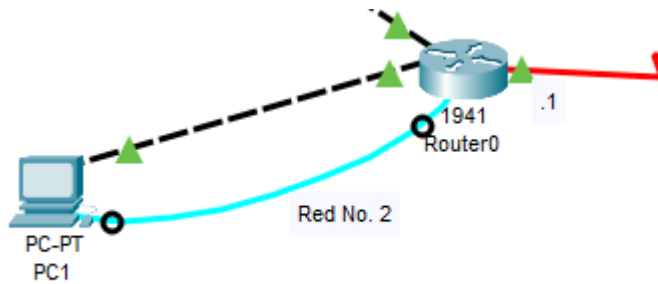
– Student 02(Julian):

- * 72.0.0.0/8
- * 73.0.0.0/8
- * 74.0.0.0/8
- * 75.0.0.0/8

Julian's Diagram



- Now, to configure it, connect from a console similarly to how you do with switches. (The blue cable is a console cable and connects to the serial port – RS232 of each device).



Configure the networks to allow addressing the specified hosts.

Networks:

For the serial interconnection network, use 20.0.0.0/8.

– Student 02(Julián):

* 72.0.0.0/8

* 73.0.0.0/8

* 74.0.0.0/8

* 75.0.0.0/8

For the serial interconnection network, use 20.0.0.0/8.

Access each router from the computer by going to the desktop tab and selecting the terminal option. Do not modify the default parameters (this is similar to what we will do going forward with the physical routers using the light blue cable at each workstation in the lab). Manually configure the router, so when prompted with "Continue with configuration dialog? [yes/no]:", select n, then enter privileged mode and proceed to configuration mode.

Student 2 (Julián)

On Router0, editing from PC using the console cable, I configured the IPs of the 3 interfaces that are being used

```
Router0(config-if)#ip address 72.0.0.1 255.0.0.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#interface gig0/1
Router0(config-if)#interface gig0/1
Router0(config-if)#ip address 73.0.0.1 255.0.0.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#interface serial0/0/0
Router0(config-if)#ip address 20.0.0.1 255.0.0.0
```

Now we do the same on router 1

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router1
Router1(config)#interface serial 0/0/0
Router1(config-if)#ip address 20.0.0.2 255.0.0.0
Router1(config-if)#clock rate 64000
Router1(config-if)#no shutdown
Router1(config-if)#exit
Router1(config)#interface gig 0/0
Router1(config-if)#ip address 74.0.0.1 255.0.0.0
Router1(config-if)#no shutdown
Router1(config-if)#exit
Router1(config)#interface gig 0/1
Router1(config-if)#ip address 75.0.0.1 255.0.0.0
Router1(config-if)#no shutdown
Router1(config-if)#exit
Router1(config)#end
Router1#
```

Now configure:

- Name of Router0: Your name.
- Name of Router1: Your surname.
- Message of day: "For exclusive use by RECO students".
- Screen synchronization.
- Block external server command lookup.
- Interface descriptions.
- Device passwords:
 - * Privilege mode: Clave E
 - * Consult: Clave C
 - * Remote terminal: Clave T

Now I modified the hostnames of the routers, changing router0 to julian and router1 to tinjaca

```
Router0>enable
Router0#configure t
Enter configuration commands, one per line.  End with CNTL/Z.
Router0(config)#hostname julian
julian(config)#
```

```
Router1#configure t
Enter configuration commands, one per line.  End with CNTL/Z.
Router1(config)#hostname tinjaca
tinjaca(config)#
```

I posted the message of the day on both routers

```
julian#config t
Enter configuration commands, one per line.  End with CNTL/Z.
julian(config)#banner motd "For exclusive use by RECO students."
julian(config)#
```

```
tinjaca#config t
Enter configuration commands, one per line.  End with CNTL/Z.
tinjaca(config)#banner motd "For exclusive use by RECO students."
tinjaca(config)#
```

To achieve screen synchronization, we enter the console line configuration, then enable synchronization and exit to return to global configuration mode.

```
julian#configure t
Enter configuration commands, one per line. End with CNTL/Z.
julian(config)#line console 0
julian(config-line)#logging synchronous
julian(config-line)#exit
julian(config)#
```

```
tinjaca#configure t
Enter configuration commands, one per line. End with CNTL/Z.
tinjaca(config)#line console 0
tinjaca(config-line)#logging synchronous
tinjaca(config-line)#exit
tinjaca(config)#
```

Now you should block the lookup of commands on the external server, that is, the automatic DNS lookup is blocked.

```
tinjaca(config)#no ip domain-lookup
tinjaca(config)#
```

```
julian(config)#no ip domain-lookup
julian(config)#
```

Now we must add a description to each interface to document its use.

```
julian(config)#interface gigabitethernet0/0
julian(config-if)#description connection to PC1
julian(config-if)#exit
julian(config)#interface gigabitethernet0/1
julian(config-if)#description connection to PC0
julian(config-if)#exit
julian(config)#interface serial0/0/0
julian(config-if)#description connection to Router(tinjaca)
julian(config-if)#exit
julian(config)#
```

```
tinjaca#configure t
Enter configuration commands, one per line. End with CNTL/Z.
tinjaca(config)#interface gigabitethernet0/0
tinjaca(config-if)#description connect to PC3
tinjaca(config-if)#exit
tinjaca(config)#interface gigabitethernet to PC2
tinjaca(config-if)#exit
^
% Invalid input detected at '^' marker.

tinjaca(config)#interface gigabitethernet to PC2
tinjaca(config-if)#exit
^
% Invalid input detected at '^' marker.

tinjaca(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

tinjaca(config)#interface gigabitethernet0/1
tinjaca(config-if)#description connection to PC2
tinjaca(config-if)#exit
tinjaca(config)#interface serial0/1/0
%Invalid interface type and number
tinjaca(config)#interface serial 0/0/0
tinjaca(config-if)#description connect to Router(julian)
tinjaca(config-if)#exit
tinjaca(config)#
```


Now we proceed to configure the passwords for the different access levels, for the privileged mode we use secret, since it allows strong encryption and for the query and remote terminal modes we use the basic command

```
Enter configuration commands, one per line. End with CNTL/Z.
julian(config)#enable secret Clave_E
julian(config)#
```

```
Enter configuration commands, one per line. End with CNTL/Z.
tinjaca(config)#enable secret Clave_E
tinjaca(config)#
```

```
julian#configure t
Enter configuration commands, one per line. End with CNTL/Z.
julian(config)#line console 0
julian(config-line)#password Clave_C
julian(config-line)#login
julian(config-line)#exit
julian(config)#end
julian#
```

```
tinjaca#configure t
Enter configuration commands, one per line. End with CNTL/Z.
tinjaca(config)#line console 0
tinjaca(config-line)#password Clave_C
tinjaca(config-line)#login
tinjaca(config-line)#exit
tinjaca(config)#end
tinjaca#
%SYS-5-CONFIG_I: Configured from console by console
```

```
julian#configure t
Enter configuration commands, one per line. End with CNTL/Z.
julian(config)#line vty 0 4
julian(config-line)#password Clave_T
julian(config-line)#login
julian(config-line)#exit
julian(config)#end
julian#
%SYS-5-CONFIG_I: Configured from console by console
```

```
tinjaca#configure t
Enter configuration commands, one per line. End with CNTL/Z.
tinjaca(config)#line vty 0 4
tinjaca(config-line)#password Clave_T
tinjaca(config-line)#login
tinjaca(config-line)#exit
tinjaca(config)#end
tinjaca#
%SYS-5-CONFIG_I: Configured from console by console
```

Finally we save the configuration so that it is maintained after a reboot.

```
julian#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
julian#
```

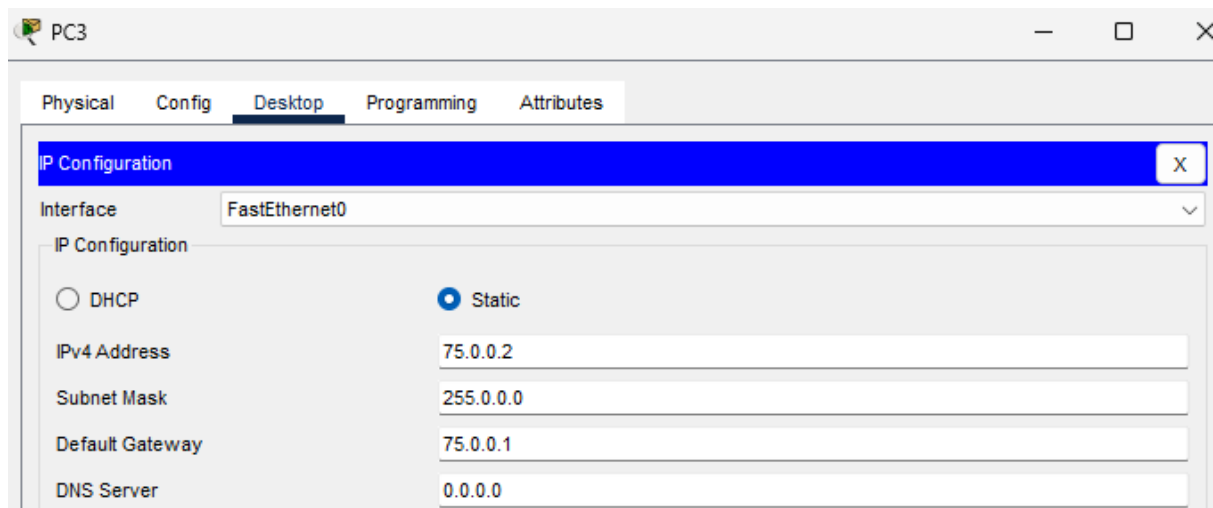
```
tinjaca#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
tinjaca#
```

Test connectivity between the PCs on the same LAN and to different networks. Which ones work and which ones don't?

First we must configure the IP, mask and gateway for each PC

The image displays three screenshots of a network configuration interface, likely from a packet tracer or similar simulation software. Each screenshot shows the 'Desktop' tab of a PC configuration window, specifically the 'IP Configuration' section. The interface includes tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Interface' is set to 'FastEthernet0'. The configuration options are set to 'Static' (selected) and 'DHCP' (unselected). The fields for 'IPv4 Address', 'Subnet Mask', 'Default Gateway', and 'DNS Server' are filled with specific values for each PC.

PC	Interface	IPv4 Address	Subnet Mask	Default Gateway	DNS Server
PC0	FastEthernet0	72.0.0.2	255.0.0.0	72.0.0.1	0.0.0.0
PC1	FastEthernet0	73.0.0.3	255.0.0.0	73.0.0.1	0.0.0.0
PC2	FastEthernet0	74.0.0.2	255.0.0.0	74.0.0.1	0.0.0.0



Now we test connectivity: Ping on the same LAN

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 73.0.0.1

Pinging 73.0.0.1 with 32 bytes of data:

Reply from 73.0.0.1: bytes=32 time<1ms TTL=255
Reply from 73.0.0.1: bytes=32 time<1ms TTL=255
Reply from 73.0.0.1: bytes=32 time<1ms TTL=255
Reply from 73.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 73.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

ping on another LAN

```
C:\>ping 74.0.0.2

Pinging 74.0.0.2 with 32 bytes of data:

Reply from 72.0.0.1: Destination host unreachable.
Reply from 72.0.0.1: Destination host unreachable.
Request timed out.
Reply from 72.0.0.1: Destination host unreachable.

Ping statistics for 74.0.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

The connection fails when trying to ping a PC located on another network.

Configure the routers with static routes so that all devices on the network can ping each other. Note: On each router, include the routes (exit interface) needed to reach networks that are not directly connected.

Every router must have static routes to reach networks not directly connected to it. If we start with the julian router, both routes from the tinjaca router are reached via the Serial 0/0/0 interface.

```
Enter configuration commands, one per line. End with CNTRL/Z.
julian(config)#ip route 74.0.0.0 255.0.0.0 serial 0/0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
julian(config)#ip route 75.0.0.0 255.0.0.0 serial 0/0/0
^
% Invalid input detected at '^' marker.

julian(config)#ip route 75.0.0.0 255.0.0.0 serial 0/0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
julian(config)#end
julian#
%SYS-5-CONFIG_I: Configured from console by console

julian#copy running-config startup-config
^
% Invalid input detected at '^' marker.

julian#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
julian#
```

And from the tinjaca router using the serial 0/1/0 interface

```
Enter configuration commands, one per line. End with CNTRL/Z.
tinjaca(config)#ip route 72.0.0.0 255.0.0.0 serial 0/1/0
%Default route without gateway, if not a point-to-point interface, may impact performance
%Invalid interface type and number
tinjaca(config)#ip route 72.0.0.0 255.0.0.0 serial 0/0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
tinjaca(config)#ip route 73.0.0.0 255.0.0.0 serial 0/0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
tinjaca(config)#end
tinjaca#
%SYS-5-CONFIG_I: Configured from console by console

tinjaca#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
tinjaca#
```

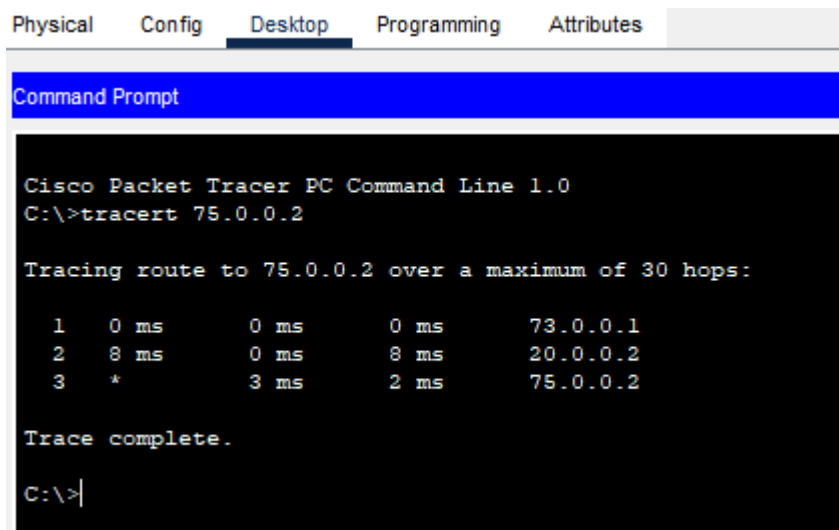
We verify that the routes have been added to the routing table

```
[OK]
julian#show ip route static
S    74.0.0.0/8 is directly connected, Serial0/0/0
S    75.0.0.0/8 is directly connected, Serial0/0/0

julian#
```

Using the ICMP traceroute command, try to verify communication between the two computers.

From PC1 to PC3



The screenshot shows the 'Desktop' tab of a PC configuration window. The 'Command Prompt' is open, displaying the output of the 'tracert 75.0.0.2' command. The output shows a successful trace with three hops: 73.0.0.1, 20.0.0.2, and 75.0.0.2.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 75.0.0.2

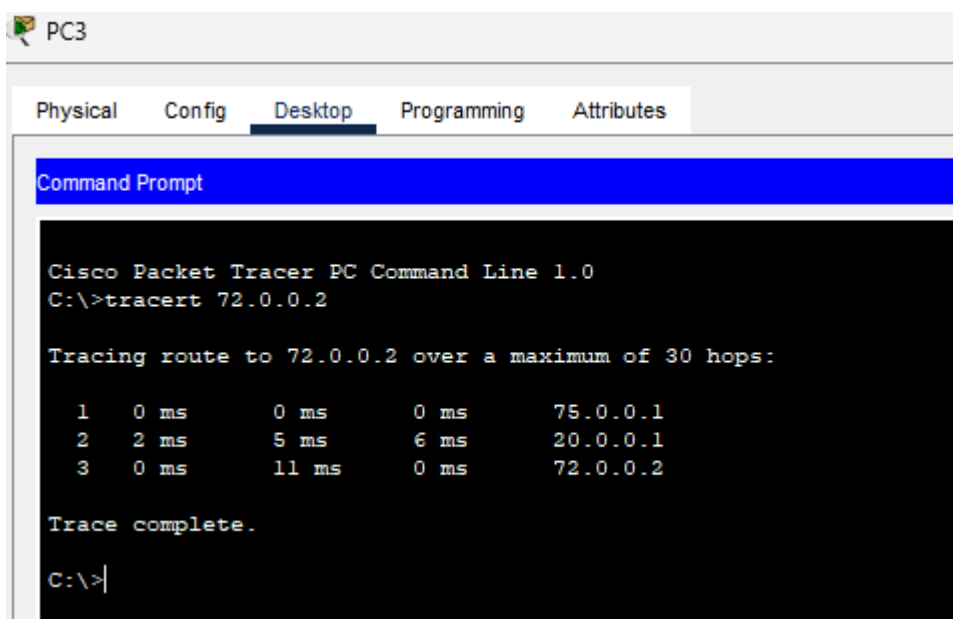
Tracing route to 75.0.0.2 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      73.0.0.1
  2  8 ms      0 ms      8 ms      20.0.0.2
  3  *         3 ms      2 ms      75.0.0.2

Trace complete.

C:\>|
```

from pc3 to pc1



The screenshot shows the 'Desktop' tab of a PC configuration window. The 'Command Prompt' is open, displaying the output of the 'tracert 72.0.0.2' command. The output shows a successful trace with three hops: 75.0.0.1, 20.0.0.1, and 72.0.0.2.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 72.0.0.2

Tracing route to 72.0.0.2 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      75.0.0.1
  2  2 ms      5 ms      6 ms      20.0.0.1
  3  0 ms      11 ms     0 ms      72.0.0.2

Trace complete.

C:\>|
```

Base Software Installation

Part of the foundational platform of an organization's IT infrastructure involves web services, which can be hosted within the company's data center or on a cloud server. These services store the organization's web pages and are used by various clients. In this lab, we will implement this service.

1. Web Service Installation

a. Install the Apache web server on the virtual machine with the Solaris operating system.

First we update the catalogs

```
root@solaris:~# pkg refresh
root@solaris:~# █
```

Now we install Apache using the command `pkg install` and verify that the package was installed correctly

```
root@solaris:~# pkg install apache-24
No hay actualizaciones para esta imagen.
root@solaris:~# pkg info | grep apache
URL de proyecto: https://apr.apache.org/
URL de origen: https://archive.apache.org/dist/apr/apr-1.6.3.tar.gz
URL de proyecto: https://apr.apache.org/
URL de origen: https://archive.apache.org/dist/apr/apr-util-1.6.1.tar.gz
Nombre: service/system/apache-stats-24
Descripción: Provides statistics about running apache instances to the
FMRI: pkg://solaris/service/system/apache-stats-24@11.4-11.4.0.0
.1.15.0:20180817T002718Z
Nombre: web/server/apache-24
FMRI: pkg://solaris/web/server/apache-24@2.4.33-11.4.0.0.1.14.0:
20180814T173250Z
URL de proyecto: https://httpd.apache.org/
URL de origen: https://archive.apache.org/dist/httpd/httpd-2.4.33.tar.gz
Nombre: web/server/apache-24/module/apache-ssl
FMRI: pkg://solaris/web/server/apache-24/module/apache-ssl@2.4.3
3-11.4.0.0.1.14.0:20180814T173244Z
URL de proyecto: https://httpd.apache.org/
URL de origen: https://archive.apache.org/dist/httpd/httpd-2.4.33.tar.gz
Nombre: web/server/apache-24/module/apache-wsgi-27
FMRI: pkg://solaris/web/server/apache-24/module/apache-wsgi-27@4
.5.1-11.4.0.0.1.14.0:20180814T173245Z
root@solaris:~# █
```

Then we searched for the service and started it.

```
root@solaris:~# svcs -a | grep -i http
disabled          9:45:03    svc:/network/http:apache24
root@solaris:~# svcadm enable svc:/network/http:apache24
root@solaris:~# █
```

b. Install the Nginx web server on the virtual machine with the Slackware Linux operating system.

First, we update the Slackware packages

```
/PACKAGES.TXT...
--2025-10-31 10:29:38-- https://mirrors.slackware.com/slackware/slackware64-15.0/pasture/PACKAGES.TXT
Resolving mirrors.slackware.com (mirrors.slackware.com)... 207.223.116.213, 64:ff9b::cfd5:74d5
Connecting to mirrors.slackware.com (mirrors.slackware.com)|207.223.116.213|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2163 (2.1K) [text/plain]
Saving to: /tmp/slackpkg.rHYOS8/pasture-PACKAGES.TXT

/tmp/slackpkg.rHYOS8/pas 100%[=====] 2.11K --.-KB/s in 0s

2025-10-31 10:29:39 (440 MB/s) - /tmp/slackpkg.rHYOS8/pasture-PACKAGES.TXT saved [2163/2163]

Downloading https://mirrors.slackware.com/slackware/slackware64-15.0/testing/PACKAGES.TXT
/PACKAGES.TXT...
--2025-10-31 10:29:39-- https://mirrors.slackware.com/slackware/slackware64-15.0/testing/PACKAGES.TXT
Resolving mirrors.slackware.com (mirrors.slackware.com)... 207.223.116.213, 64:ff9b::cfd5:74d5
Connecting to mirrors.slackware.com (mirrors.slackware.com)|207.223.116.213|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://mirror.cedia.org.ec/slackware/slackware64-15.0/testing/PACKAGES.TXT [following]
--2025-10-31 10:29:40-- https://mirror.cedia.org.ec/slackware/slackware64-15.0/testing/PACKAGES.TXT
Resolving mirror.cedia.org.ec (mirror.cedia.org.ec)... 201.159.221.67, 64:ff9b::c99f:dd43
Connecting to mirror.cedia.org.ec (mirror.cedia.org.ec)|201.159.221.67|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 20561 (20K) [text/plain]
Saving to: /tmp/slackpkg.rHYOS8/testing-PACKAGES.TXT

/tmp/slackpkg.rHYOS8/tes 100%[=====] 20.08K --.-KB/s in 0s

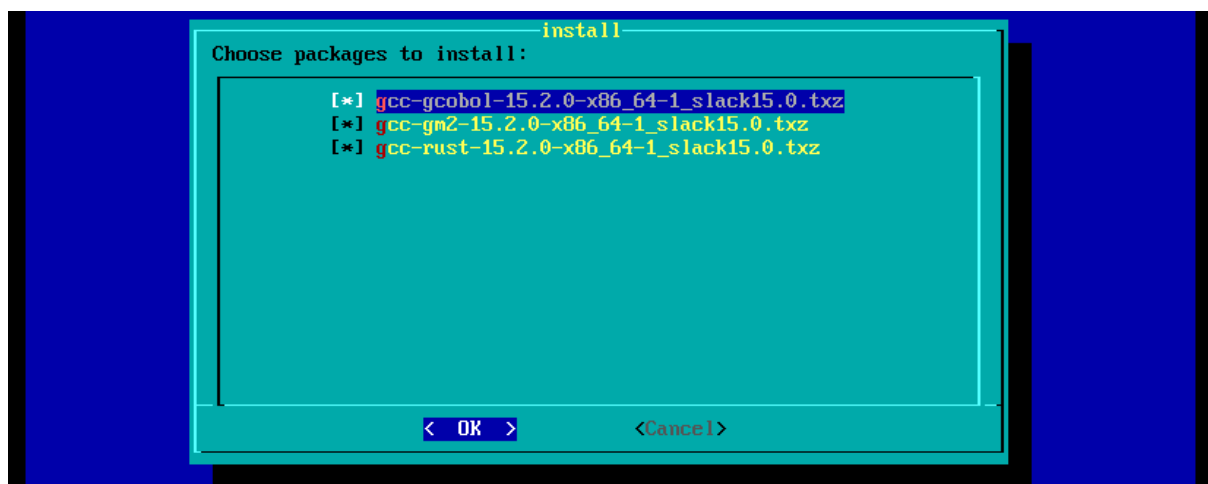
2025-10-31 10:29:40 (157 MB/s) - /tmp/slackpkg.rHYOS8/testing-PACKAGES.TXT saved [20561/20561]

Formatting lists to slackpkg style...
Package List: using CHECKSUMS.md5 as source
Package descriptions

root@aysrSlack:~# _
```

Now we install some compilation-related dependencies

```
root@aysrSlack:~# slackpkg install gcc make pcre-devel zlib-devel openssl-devel_
```



```
#
Executing install script for gcc-gcobol-15.2.0-x86_64-1_slack15.0.txz.
Package gcc-gcobol-15.2.0-x86_64-1_slack15.0.txz installed.
    Package gcc-gm2-15.2.0-x86_64-1_slack15.0.txz is already in cache - not downloading
    Installing gcc-gm2-15.2.0-x86_64-1_slack15.0...
Verifying package gcc-gm2-15.2.0-x86_64-1_slack15.0.txz.
Installing package gcc-gm2-15.2.0-x86_64-1_slack15.0.txz:
PACKAGE DESCRIPTION:
# gcc-gm2 (Modula-2 support for GCC)
#
# Modula-2 support for the GNU Compiler Collection.
#
# Modula-2 language support covering ISO/IEC 10514-1, PIM2, PIM3, and
# PIM4 dialects together with a complete set of ISO/IEC 10514-1 and PIM
# libraries.
#
Executing install script for gcc-gm2-15.2.0-x86_64-1_slack15.0.txz.
Package gcc-gm2-15.2.0-x86_64-1_slack15.0.txz installed.
    Package gcc-rust-15.2.0-x86_64-1_slack15.0.txz is already in cache - not downloading
    Installing gcc-rust-15.2.0-x86_64-1_slack15.0...
Verifying package gcc-rust-15.2.0-x86_64-1_slack15.0.txz.
Installing package gcc-rust-15.2.0-x86_64-1_slack15.0.txz:
PACKAGE DESCRIPTION:
# gcc-rust (Rust support for GCC)
#
# Rust support for the GNU Compiler Collection.
#
# gccrs is a full alternative implementation of the Rust language ontop
# of GCC with the goal to become fully upstream with the GNU toolchain.
# Please note, the compiler is in a very early stage and not usable yet
# for compiling real Rust programs.
#
Package gcc-rust-15.2.0-x86_64-1_slack15.0.txz installed.
Searching for NEW configuration files...
    No .new files found.

root@aysrSlack:~#
```

Now we download and compile nginx

```
root@aysrSlack:/usr/local/src# wget http://nginx.org/download/nginx-1.26.1.tar.gz
--2025-10-31 10:55:17-- http://nginx.org/download/nginx-1.26.1.tar.gz
Resolving nginx.org (nginx.org)... 52.58.199.22, 3.125.197.172, 64:ff9b::343a:c716, ...
Connecting to nginx.org (nginx.org)|52.58.199.22|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1244738 (1.2M) [application/octet-stream]
Saving to: 'nginx-1.26.1.tar.gz'

nginx-1.26.1.tar.gz      100%[=====>] 1.19M  501KB/s   in 2.4s

2025-10-31 10:55:20 (501 KB/s) - 'nginx-1.26.1.tar.gz' saved [1244738/1244738]

root@aysrSlack:/usr/local/src#
```

```
root@aysrSlack:/usr/local/src# tar zxvf nginx-*.tar.gz_
```

```
nginx-1.26.1/auto/lib/libxslt/
nginx-1.26.1/auto/lib/perl/
nginx-1.26.1/auto/lib/geoip/
nginx-1.26.1/auto/lib/zlib/
nginx-1.26.1/auto/lib/google-perftools/
nginx-1.26.1/auto/lib/make
nginx-1.26.1/auto/lib/pcre/
nginx-1.26.1/auto/lib/openssl/
nginx-1.26.1/auto/lib/conf
nginx-1.26.1/auto/lib/openssl/makefile.bcc
nginx-1.26.1/auto/lib/openssl/makefile.msuc
nginx-1.26.1/auto/lib/openssl/make
nginx-1.26.1/auto/lib/openssl/conf
nginx-1.26.1/auto/lib/pcre/makefile.bcc
nginx-1.26.1/auto/lib/pcre/makefile.msuc
nginx-1.26.1/auto/lib/pcre/make
nginx-1.26.1/auto/lib/pcre/conf
```



```
nginx-1.26.1/auto/lib/pcre/makefile.owc
nginx-1.26.1/auto/lib/google-perftools/conf
nginx-1.26.1/auto/lib/zlib/makefile.bcc
nginx-1.26.1/auto/lib/zlib/makefile.msuc
nginx-1.26.1/auto/lib/zlib/make
nginx-1.26.1/auto/lib/zlib/conf
nginx-1.26.1/auto/lib/zlib/makefile.owc
nginx-1.26.1/auto/lib/geoip/conf
nginx-1.26.1/auto/lib/perl/make
nginx-1.26.1/auto/lib/perl/conf
nginx-1.26.1/auto/lib/libxslt/conf
nginx-1.26.1/auto/lib/libatomic/make
nginx-1.26.1/auto/lib/libatomic/conf
nginx-1.26.1/auto/lib/libgd/conf
nginx-1.26.1/auto/types/typedef
nginx-1.26.1/auto/types/value
nginx-1.26.1/auto/types/uintptr_t
nginx-1.26.1/auto/types/sizeof
nginx-1.26.1/man/nginx.8
root@aysrSlack:/usr/local/src#
```

```
root@aysrSlack:/usr/local/src# cd nginx-*/
root@aysrSlack:/usr/local/src/nginx-1.26.1#
```

```
root@aysrSlack:/usr/local/src/nginx-1.26.1# ./configure --sbin-path=/usr/sbin/nginx --conf-path=/etc/nginx/nginx.conf --prefix=/usr/local/nginx --pid-path=/var/run/nginx.pid --with-http_ssl_module --with-http_v2_module_
```

```
checking for POSIX semaphores ... not found
checking for POSIX semaphores in libpthread ... found
checking for struct msghdr.msg_control ... found
checking for ioctl(FIONBIO) ... found
checking for ioctl(FIONREAD) ... found
checking for struct tm.tm_gmtoff ... found
checking for struct dirent.d_namlen ... not found
checking for struct dirent.d_type ... found
checking for sysconf(_SC_NPROCESSORS_ONLN) ... found
checking for sysconf(_SC_LEVEL1_DCACHE_LINESIZE) ... found
checking for openat(), fstatat() ... found
checking for getaddrinfo() ... found
checking for PCRE2 library ... found
checking for OpenSSL library ... found
checking for zlib library ... found
creating objs/Makefile
```

Configuration summary

- + using system PCRE2 library
- + using system OpenSSL library
- + using system zlib library

```
nginx path prefix: "/usr/local/nginx"
nginx binary file: "/usr/sbin/nginx"
nginx modules path: "/usr/local/nginx/modules"
nginx configuration prefix: "/etc/nginx"
nginx configuration file: "/etc/nginx/nginx.conf"
nginx pid file: "/var/run/nginx.pid"
nginx error log file: "/usr/local/nginx/logs/error.log"
nginx http access log file: "/usr/local/nginx/logs/access.log"
nginx http client request body temporary files: "client_body_temp"
nginx http proxy temporary files: "proxy_temp"
nginx http fastcgi temporary files: "fastcgi_temp"
nginx http uwsgi temporary files: "uwsgi_temp"
nginx http scgi temporary files: "scgi_temp"
```

```
root@aysrSlack:/usr/local/src/nginx-1.26.1# _
```

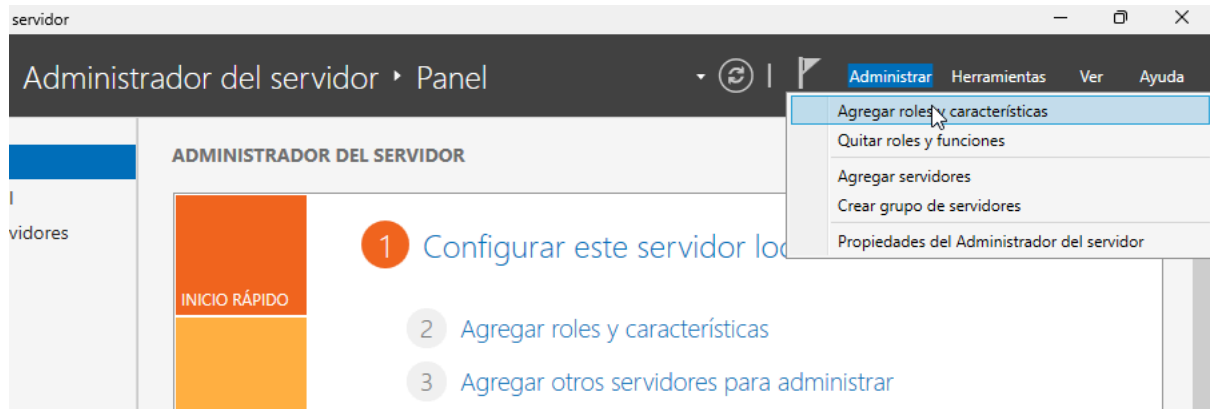
Then use make and make install commands

```
objs/src/http/modules/nginx_http_index_module.o \
objs/src/http/modules/nginx_http_mirror_module.o \
objs/src/http/modules/nginx_http_try_files_module.o \
objs/src/http/modules/nginx_http_auth_basic_module.o \
objs/src/http/modules/nginx_http_access_module.o \
objs/src/http/modules/nginx_http_limit_conn_module.o \
objs/src/http/modules/nginx_http_limit_req_module.o \
objs/src/http/modules/nginx_http_geo_module.o \
objs/src/http/modules/nginx_http_map_module.o \
objs/src/http/modules/nginx_http_split_clients_module.o \
objs/src/http/modules/nginx_http_referer_module.o \
objs/src/http/modules/nginx_http_rewrite_module.o \
objs/src/http/modules/nginx_http_ssl_module.o \
objs/src/http/modules/nginx_http_proxy_module.o \
objs/src/http/modules/nginx_http_fastcgi_module.o \
objs/src/http/modules/nginx_http_uwsgi_module.o \
objs/src/http/modules/nginx_http_scgi_module.o \
objs/src/http/modules/nginx_http_grpc_module.o \
objs/src/http/modules/nginx_http_memcached_module.o \
objs/src/http/modules/nginx_http_empty_gif_module.o \
objs/src/http/modules/nginx_http_browser_module.o \
objs/src/http/modules/nginx_http_upstream_hash_module.o \
objs/src/http/modules/nginx_http_upstream_ip_hash_module.o \
objs/src/http/modules/nginx_http_upstream_least_conn_module.o \
objs/src/http/modules/nginx_http_upstream_random_module.o \
objs/src/http/modules/nginx_http_upstream_keepalive_module.o \
objs/src/http/modules/nginx_http_upstream_zone_module.o \
objs/nginx_modules.o \
-lldl -lpthread -lcrypt -lpcrc2-8 -lssl -lcrypto -ldl -lpthread -lz \
-Wl,-E
sed -e "s|%%PREFIX%%|/usr/local/nginx|" \
-e "s|%%PID_PATH%%|/var/run/nginx.pid|" \
-e "s|%%CONF_PATH%%|/etc/nginx/nginx.conf|" \
-e "s|%%ERROR_LOG_PATH%%|/usr/local/nginx/logs/error.log|" \
< man/nginx.8 > objs/nginx.8
make[1]: Leaving directory '/usr/local/src/nginx-1.26.1'
root@aysrSlack:/usr/local/src/nginx-1.26.1#
```

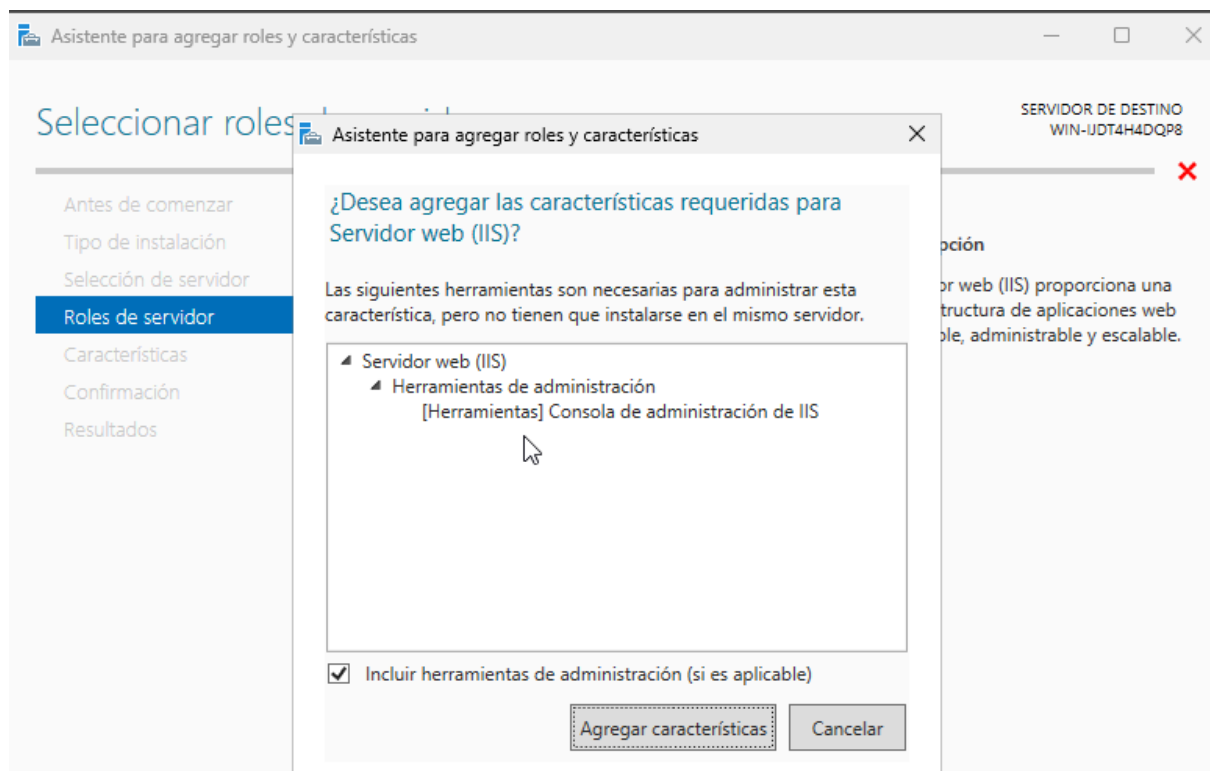
```
cp conf/mime.types '/etc/nginx/mime.types.default'
test -f '/etc/nginx/fastcgi_params' \
|| cp conf/fastcgi_params '/etc/nginx'
cp conf/fastcgi_params \
'/etc/nginx/fastcgi_params.default'
test -f '/etc/nginx/fastcgi.conf' \
|| cp conf/fastcgi.conf '/etc/nginx'
cp conf/fastcgi.conf '/etc/nginx/fastcgi.conf.default'
test -f '/etc/nginx/uwsgi_params' \
|| cp conf/uwsgi_params '/etc/nginx'
cp conf/uwsgi_params \
'/etc/nginx/uwsgi_params.default'
test -f '/etc/nginx/scgi_params' \
|| cp conf/scgi_params '/etc/nginx'
cp conf/scgi_params \
'/etc/nginx/scgi_params.default'
test -f '/etc/nginx/nginx.conf' \
|| cp conf/nginx.conf '/etc/nginx/nginx.conf'
cp conf/nginx.conf '/etc/nginx/nginx.conf.default'
test -d '/var/run' \
|| mkdir -p '/var/run'
test -d '/usr/local/nginx/logs' \
|| mkdir -p '/usr/local/nginx/logs'
test -d '/usr/local/nginx/html' \
|| cp -R html '/usr/local/nginx'
test -d '/usr/local/nginx/logs' \
|| mkdir -p '/usr/local/nginx/logs'
make[1]: Leaving directory '/usr/local/src/nginx-1.26.1'
root@aysrSlack:/usr/local/src/nginx-1.26.1#
```

c. Configure the web server that comes with Windows Server.

First we open Server Manager → Manage → Add Roles and Features.



In server roles, we select IIS web server.



d. Create a simple webpage on each server to test that it works.

On Oracle Solaris

Now we create the new page

```
root@solaris:~# mkdir -p /var/www/html
root@solaris:~# echo "<html><body><h1>Apache en Solaris - OK</h1></body></html>" | tee /var/www/html/index.html
<html><body><h1>Apache en Solaris - OK</h1></body></html>
root@solaris:~#
```

```

root@solaris:~# chown -R root:root /var/www/html
root@solaris:~# chmod -R 755 /var/www/html
root@solaris:~# █

```

Then verify the connection

```

root@solaris:~# curl -I http://localhost
HTTP/1.1 200 OK
Date: Fri, 31 Oct 2025 15:14:11 GMT
Server: Apache/2.4.33 (Unix)
Last-Modified: Fri, 17 Aug 2018 02:07:27 GMT
ETag: "2d-57398056ff5c0"
Accept-Ranges: bytes
Content-Length: 45
Content-Type: text/html

root@solaris:~# curl http://localhost
<html><body><h1>It works!</h1></body></html>
root@solaris:~# █

```

On Linux Slackware

Now we configure the test page and the nginx configuration

```

root@ayrsSlack:/usr/local# mkdir -p /usr/local/nginx/html
root@ayrsSlack:/usr/local# echo "<html><body><h1>Nginx en Slackware - OK</h1></body></html>" | tee /
usr/local/nginx/html/index.html
<html><body><h1>Nginx en Slackware - OK</h1></body></html>
root@ayrsSlack:/usr/local#

```

```

GNU nano 6.0 /etc/nginx/nginx.conf Modified
user nobody;
worker_processes 1;

#error_log logs/error.log;
#error_log logs/error.log notice;
#error_log logs/error.log info;
error_log /var/log/nginx/error.log warn;
#pid logs/nginx.pid;
pid /var/run/nginx.pid;

events {
    worker_connections 1024;
}

http {
    include mime.types;
    default_type application/octet-stream;

    #log_format main '$remote_addr - $remote_user [$time_local] "$request" '
    # '$status $body_bytes_sent "$http_referer" '
    # '"$http_user_agent" "$http_x_forwarded_for"';

    #access_log logs/access.log main;

    sendfile on;
    #tcp_nopush on;

    #keepalive_timeout 0;
    keepalive_timeout 65;

    #gzip on;

    ^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo
    ^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_ Go To Line  ^E Redo

```

```
GNU nano 6.0 /etc/nginx/nginx.conf Modified
server {
    listen      80;
    server_name localhost;

    #charset koi8-r;

    #access_log logs/host.access.log main;

    location / {
        root    /usr/local/nginx/html;
        index   index.html index.htm;
    }

    #error_page 404          /404.html;

    # redirect server error pages to the static page /50x.html
    #
    error_page 500 502 503 504 /50x.html;
    location = /50x.html {
        root    html;
    }

    # proxy the PHP scripts to Apache listening on 127.0.0.1:80
    #
    #location ~ \.php$ {
    #    proxy_pass http://127.0.0.1;
    #}

    # pass the PHP scripts to FastCGI server listening on 127.0.0.1:9000
    #
    #location ~ \.php$ {
    #    root           html;
    #    fastcgi_pass   127.0.0.1:9000;
    #}
```

Now start Nginx

```
root@aysrSlack:/usr/local# mkdir -p /var/log/nginx
root@aysrSlack:/usr/local# touch /var/log/nginx/error.log
root@aysrSlack:/usr/local# chown -R nobody:nogroup /var/log/nginx
root@aysrSlack:/usr/local#
```

In this case, the httpd process had to be stopped because it was using port 80.

```
root@aysrSlack:/usr/local# /usr/sbin/nginx
nginx: [emerg] open() "/var/log/nginx/error.log.warn" failed (2: No such file or directory)
root@aysrSlack:/usr/local# mkdir -p /var/log/nginx
root@aysrSlack:/usr/local# touch /var/log/nginx/error.log
root@aysrSlack:/usr/local# chown -R nobody:nogroup /var/log/nginx
root@aysrSlack:/usr/local# ls -l /var/log/nginx
total 0
-rw-r--r-- 1 nobody nogroup 0 Oct 31 12:04 error.log
root@aysrSlack:/usr/local# /usr/sbin/nginx
nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
nginx: [emerg] still could not bind()
root@aysrSlack:/usr/local# netstat -tulnp | grep :80
tcp6      0      0 :::80                :::*                  LISTEN    1332/httpd
root@aysrSlack:/usr/local# /etc/rc.d/rc.httpd stop
root@aysrSlack:/usr/local#
```

Now we can start Nginx

```
root@aysrSlack:/usr/local# /usr/sbin/nginx
root@aysrSlack:/usr/local# /usr/sbin/nginx -s quit
root@aysrSlack:/usr/local#
```

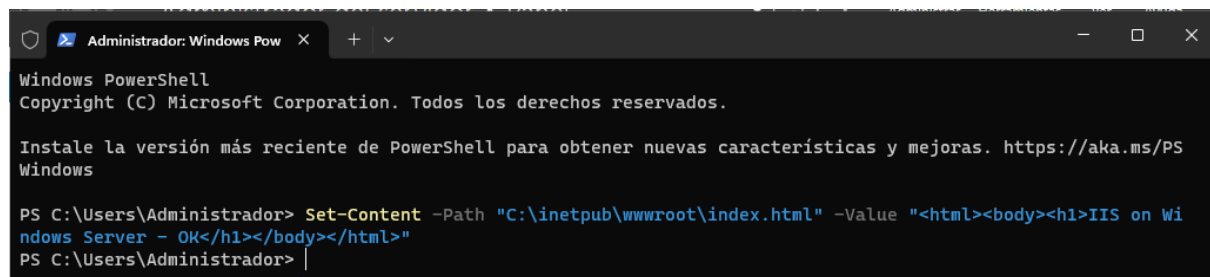
Then, prove that the server is running correctly

```
root@aysrSlack:~# /usr/sbin/nginx
root@aysrSlack:~# netstat -tulnp | grep :80
tcp        0      0 0.0.0.0:80          0.0.0.0:*          LISTEN     1489/nginx: master
root@aysrSlack:~# _

root@aysrSlack:/usr/local# curl http://localhost
<html><body><h1>Nginx en Slackware - OK</h1></body></html>
root@aysrSlack:/usr/local#
```

On Windows Server

Now we create a simple web page to test the server's functionality.

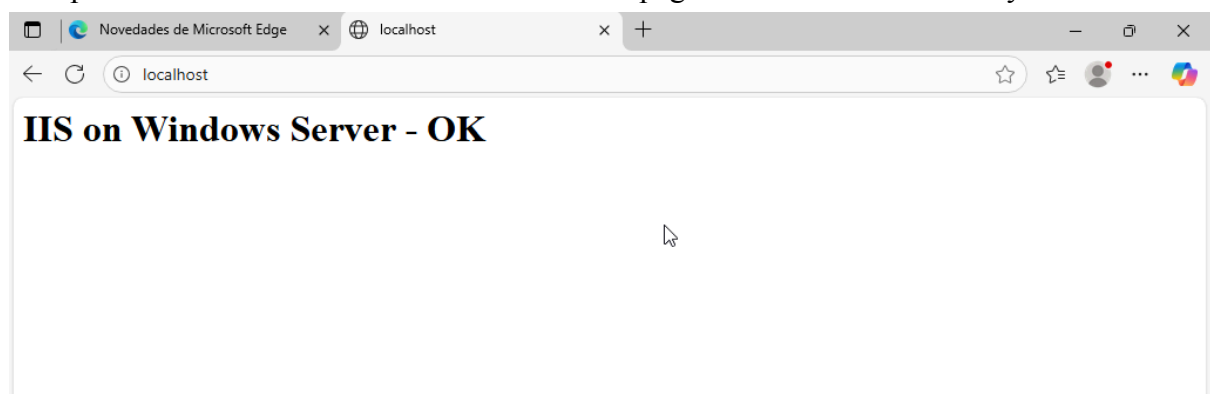


```
Administrador: Windows Pow x + -
Windows PowerShell
Copyright (C) Microsoft Corporation. Todos los derechos reservados.

Instale la versión más reciente de PowerShell para obtener nuevas características y mejoras. https://aka.ms/PSWindows

PS C:\Users\Administrador> Set-Content -Path "C:\inetpub\wwwroot\index.html" -Value "<html><body><h1>IIS on Windows Server - OK</h1></body></html>"
PS C:\Users\Administrador>
```

We opened the browser and verified that the web page was created successfully.



e. Configure the web servers to start automatically when the operating system boots.

On Oracle Solaris

We check if the Apache service is already registered in SMF; if it is online, it means it is already configured to start automatically. Otherwise, we type “svcadm enable svc:/network/http:apache24”

```
root@solaris:~# svcs -a | grep -i apache
disabled    10:03:54    svc:/system/apache-stats-24:default
online      10:04:20    svc:/network/http:apache24
root@solaris:~# Nov  4 10:05:27 solaris sendmail[945]: unable to qualify my own
```

On Linux Slackware

We created the following script to configure startup

```
GNU nano 6.0 /etc/rc.d/rc.nginx Modified
#!/bin/sh

/etc/rc.d/rc.httpd stop

case "$1" in
    start)
        echo "Starting Nginx..."
        /usr/sbin/nginx
        ;;
    stop)
        echo "Stopping Nginx..."
        /usr/sbin/nginx -s quit
        ;;
    restart)
        echo "Restarting Nginx..."
        /usr/sbin/nginx -s quit
        sleep 1
        /usr/sbin/nginx
        ;;
    *)
        echo "Use: $0 {start|stop|restart}"
        exit 1
esac
```

we give administrator permissions

```
root@aysrSlack:~# chmod +x /etc/rc.d/rc.nginx
root@aysrSlack:~#
```

We tested that the script works correctly

```
root@aysrSlack:~# /etc/rc.d/rc.nginx start
Starting Nginx...
root@aysrSlack:~# _

root@aysrSlack:~# ps aux | grep nginx
root      1532  0.0  0.0  8488  800 ?        Ss   10:26   0:00 nginx: master process /usr/sbin/ng
inx
nobody    1533  0.0  0.1  8944  2744 ?        S    10:26   0:00 nginx: worker process
root      1545  0.0  0.1  3984  2144 tty1    S+   10:29   0:00 grep nginx
root@aysrSlack:~# _
```

To make it run when the machine starts, we add this to the script.

```
if [ -x /etc/rc.d/rc.nginx ] ; then
    /etc/rc.d/rc.nginx start
fi_
```

After restarting the machine, we run the following command to verify that nginx starts successfully.

```
Welcome to Linux 5.15.19 x86_64 (tty1)

aysrSlack login: root
Password:
Last login: Tue Nov  4 10:13:39 on tty1
Linux 5.15.19.

No self-respecting fish would want to be wrapped in that kind of paper.
    -- Mike Royko on the Chicago Sun-Times after it was
        taken over by Rupert Murdoch

root@aysrSlack:~# ps aux | grep nginx
root      1482  0.0  0.1  3984  2104 tty1      S+   10:38   0:00 grep nginx
root@aysrSlack:~# _
```

On Windows Server

Normally, the Windows Server service is configured to start automatically when the machine starts, so we simply turn on the machine and verify that it is running.

```
PS C:\Users\Administrador> Get-Service -Name W3SVC

Status      Name      DisplayName
-----
Running     W3SVC     Servicio de publicación World Wide Web

PS C:\Users\Administrador> |
```

f. From another computer, test access to the web servers.

```
Command Prompt

(c) Microsoft Corporation. All rights reserved.

C:\Users\TEMP.LABINFO>curl http://10.2.77.35
curl: (28) Failed to connect to 10.2.77.35 port 80 after 21073 ms: Could not connect to server

C:\Users\TEMP.LABINFO>curl http://10.2.77.31
<html><body><h1>It works!</h1></body></html>

C:\Users\TEMP.LABINFO>curl http://10.2.77.35
<html><body><h1>It works!</h1></body></html>

C:\Users\TEMP.LABINFO>http://10.2.77.33
'http:' is not recognized as an internal or external command,
operable program or batch file.

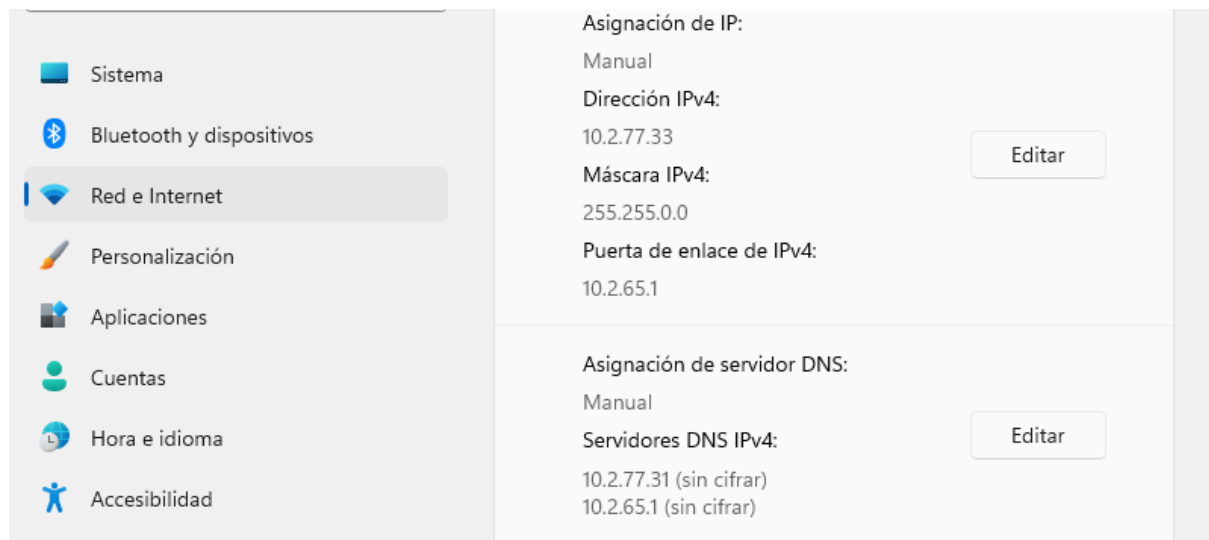
C:\Users\TEMP.LABINFO>curl http://10.2.77.33
<html><body><h1>IIS on Windows Server - OK</h1></body></html>

C:\Users\TEMP.LABINFO>
```

g. Configure the DNS service so that each web server can be accessed by name.

Since we have BIND configured on the machines, we can simply configure DNS on each machine and enter the domain names along with their IP addresses.

We configured Windows Server DNS to point to the master server, which will be running Linux Slackware.



On Oracle Solaris, we use the DNS configuration in the lab03 and we add this in the .hosts file

```
ostrich.santiago.com.it.      IN      A       10.2.77.34
;
www.santiago.com.it.         IN      A       10.2.77.32
auyamas.santiago.com.it.     IN      AAAAA  2001::1
momia.santiago.com.it.       IN      AAAAA  2001::2
;
; ===== NEW WEB SERVERS =====
;
apache.santiago.com.it.      IN      A       10.2.77.35
nginx.santiago.com.it.       IN      A       10.2.77.31
iis.santiago.com.it.         IN      A       10.2.77.33
;
; =====
;
; Aliases
;
red.santiago.com.it.         IN      CNAME   www.santiago.com.it.
```

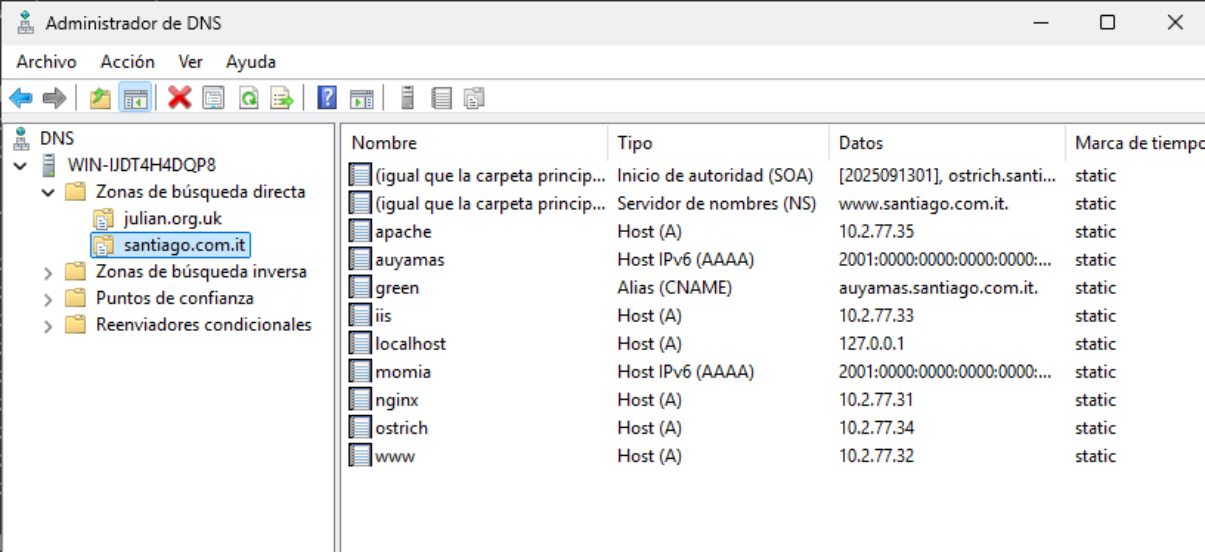
We verify that the file has no errors after editing it.

```
root@solaris:~# named-checkzone santiago.com.it /etc/DNS/santiago.com.it.hosts
/etc/DNS/santiago.com.it.hosts:6: no TTL specified; using SOA MINTTL instead
zone santiago.com.it/IN: loaded serial 2025091301
OK
root@solaris:~#
```

And restart de DNS service

```
root@solaris:~# svcadm restart svc:/network/dns/server:default
root@solaris:~# svcadm restart svc:/network/dns/client:default
root@solaris:~#
```

We refreshed the slave server in Windows Server so that the new domain names appear.



The screenshot shows the 'Administrador de DNS' window. On the left, the tree view shows 'DNS' > 'WIN-IJD4H4DQP8' > 'Zonas de búsqueda directa' > 'santiago.com.it'. The main pane displays a table of DNS records for this zone.

Nombre	Tipo	Datos	Marca de tiempo
(igual que la carpeta princip...	Inicio de autoridad (SOA)	[2025091301], ostrich.santi...	static
(igual que la carpeta princip...	Servidor de nombres (NS)	www.santiago.com.it.	static
apache	Host (A)	10.2.77.35	static
auyamas	Host IPv6 (AAAA)	2001:0000:0000:0000:0000:...	static
green	Alias (CNAME)	auyamas.santiago.com.it.	static
iis	Host (A)	10.2.77.33	static
localhost	Host (A)	127.0.0.1	static
momia	Host IPv6 (AAAA)	2001:0000:0000:0000:0000:...	static
nginx	Host (A)	10.2.77.31	static
ostrich	Host (A)	10.2.77.34	static
www	Host (A)	10.2.77.32	static

And prove that its functionality

```
C:\Users\Administrador>nslookup nginx.santiago.com.it
Servidor: UnKnown
Address: 10.2.77.35

Nombre: nginx.santiago.com.it
Address: 10.2.77.31

C:\Users\Administrador>nslookup apache.santiago.com.it
Servidor: UnKnown
Address: 10.2.77.35

Nombre: apache.santiago.com.it
Address: 10.2.77.35

C:\Users\Administrador>nslookup iis.santiago.com.it
Servidor: UnKnown
Address: 10.2.77.35

Nombre: iis.santiago.com.it
Address: 10.2.77.33

C:\Users\Administrador>
```

Then, we can search by domain name



The screenshot shows a 'Command Prompt' window with the following commands and output:

```
C:\Users\Administrador>curl http://apache.santiago.com.it
<html><body><h1>It works!</h1></body></html>

C:\Users\Administrador>curl http://iis.santiago.com.it
<html><body><h1>IIS on Windows Server - OK</h1></body></html>

C:\Users\Administrador>
```

2. Web Service Application

a. Enable the PHP extension for the Apache Server

First we install php-71

```
root@solaris:~# pkg install pkg://solaris/web/php-71
Paquetes que instalar: 5
Mediadores que cambiar: 1
Crear entorno de inicio: No
Crear copia de seguridad de entorno de inicio: No

DOWNLOAD                                PKGS      FILES      XFER (MB)   SPEED
Completado                             5/5       783/783     44.8/44.8   749k/s

PHASE                                ITEMS
Instalando acciones nuevas              939/939
Actualizando base de datos de estado de paquete      Listo
Actualizando cachÃ© de paquete                0/0
Actualizando estado de imagen                Listo
Creando base de datos de bÃºsqueda rÃ¡pida          Listo
Actualizando cachÃ© de paquete                1/1
root@solaris:~# █
```

Next, we enable PHP in Apache, and then we stop Apache.

```
root@solaris:~# svcadm disable svc:/network/http:apache24
root@solaris:~# █
```

We downloaded the pre-compiled package directly from a PostgreSQL mirror

```
root@solaris:~# cd /tmp
root@solaris:/tmp# wget https://ftp.postgresql.org/pub/source/v13.3/postgresql-13.3.tar.gz
--2025-11-07 12:16:08-- https://ftp.postgresql.org/pub/source/v13.3/postgresql-13.3.tar.gz
Resolving ftp.postgresql.org (ftp.postgresql.org)... 199.232.179.52, 2a04:4e42:6c::820
Connecting to ftp.postgresql.org (ftp.postgresql.org)|199.232.179.52|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 27610951 (26M) [application/gzip]
Saving to: 'postgresql-13.3.tar.gz'

postgresql-13.3.tar.gz  100%[=====>] 26.33M  3.48MB/s  in 7.5s

2025-11-07 12:16:17 (3.50 MB/s) - 'postgresql-13.3.tar.gz' saved [27610951/27610951]
root@solaris:/tmp# █
```

We unzipped the package and also installed gcc so we could compile.

```

root@solaris:/tmp# tar -xzf postgresql-13.3.tar.gz
root@solaris:/tmp# cd postgresql-13.3
root@solaris:/tmp/postgresql-13.3# ./configure --prefix=/usr/local/pgsql
checking build system type... i386-pc-solaris2.11
checking host system type... i386-pc-solaris2.11
checking which template to use... solaris
checking whether NLS is wanted... no
checking for default port number... 5432
checking for block size... 8kB
checking for segment size... 1GB
checking for WAL block size... 8kB
checking for gcc... no
checking for cc... no
configure: error: in '/tmp/postgresql-13.3':
configure: error: no acceptable C compiler found in $PATH
See 'config.log' for more details
root@solaris:/tmp/postgresql-13.3# pkg install developer/gcc-11
No updates necessary for this image.
root@solaris:/tmp/postgresql-13.3# gcc --version
-bash: gcc: command not found
root@solaris:/tmp/postgresql-13.3# pkg install developer/gcc
    Packages to install: 16
    Mediators to change: 1
    Services to change: 1
    Create boot environment: No
    Create backup boot environment: No

DOWNLOAD                                PKGS      FILES    XFER (MB)   SPEED
Completed                             16/16     1834/1834  332.4/332.4  1.6M/s

PHASE                                ITEMS
Installing new actions                2298/2298
Updating package state database         Done
Updating package cache                 0/0
Updating image state                   Done
Creating fast lookup database          Done
Updating package cache                 1/1
root@solaris:/tmp/postgresql-13.3# |

```

We compile and install use gmake and gmake install

```

root@solaris:/tmp/postgresql-13.3# ./configure --prefix=/usr/local/pgsql
checking build system type... x86_64-pc-solaris2.11
checking host system type... x86_64-pc-solaris2.11
checking which template to use... solaris
checking whether NLS is wanted... no
checking for default port number... 5432
checking for block size... 8kB
checking for segment size... 1GB
checking for WAL block size... 8kB
checking for gcc... gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... no
checking for suffix of object files... o
checking whether we are using the GNU C compiler... yes
checking whether gcc accepts -g... yes
checking for gcc option to accept ISO C89... none needed
checking for gcc option to accept ISO C99... none needed
checking for g++... g++
checking whether we are using the GNU C++ compiler... yes
checking whether g++ accepts -g... yes
checking for gawk... gawk
checking whether gcc supports -Wdeclaration-after-statement, for CFLAGS... yes
checking whether gcc supports -Werror=vla, for CFLAGS... yes
checking whether gcc supports -Wendif-labels, for CFLAGS... yes
checking whether g++ supports -Wendif-labels, for CXXFLAGS... yes
checking whether gcc supports -Wmissing-format-attribute, for CFLAGS... yes
checking whether g++ supports -Wmissing-format-attribute, for CXXFLAGS... yes
checking whether gcc supports -Wimplicit-fallthrough=3, for CFLAGS... yes
checking whether g++ supports -Wimplicit-fallthrough=3, for CXXFLAGS... yes
checking whether gcc supports -Wformat-security, for CFLAGS... yes

```

enter gmake

```
gmake[2]: Leaving directory '/tmp/postgresql-13.3/src/test/isolation'
gmake -C test/perl all
gmake[2]: Entering directory '/tmp/postgresql-13.3/src/test/perl'
gmake[2]: Nothing to be done for 'all'.
gmake[2]: Leaving directory '/tmp/postgresql-13.3/src/test/perl'
gmake[1]: Leaving directory '/tmp/postgresql-13.3/src'
gmake -C config all
gmake[1]: Entering directory '/tmp/postgresql-13.3/config'
gmake[1]: Nothing to be done for 'all'.
gmake[1]: Leaving directory '/tmp/postgresql-13.3/config'
All of PostgreSQL successfully made. Ready to install.
root@solaris:/tmp/postgresql-13.3#
```

now gmake install

```
/usr/bin/gmkdir -p '/usr/local/pgsql/lib/pgxs/src'
/usr/bin/ginstall -c -m 644 Makefile.global '/usr/local/pgsql/lib/pgxs/src/Makefile.global'
/usr/bin/ginstall -c -m 644 Makefile.port '/usr/local/pgsql/lib/pgxs/src/Makefile.port'
/usr/bin/ginstall -c -m 644 ./Makefile.shlib '/usr/local/pgsql/lib/pgxs/src/Makefile.shlib'
/usr/bin/ginstall -c -m 644 ./nls-global.mk '/usr/local/pgsql/lib/pgxs/src/nls-global.mk'
gmake[1]: Leaving directory '/tmp/postgresql-13.3/src'
gmake -C config install
gmake[1]: Entering directory '/tmp/postgresql-13.3/config'
/usr/bin/gmkdir -p '/usr/local/pgsql/lib/pgxs/config'
/usr/bin/ginstall -c -m 755 ./install-sh '/usr/local/pgsql/lib/pgxs/config/install-sh'
/usr/bin/ginstall -c -m 755 ./missing '/usr/local/pgsql/lib/pgxs/config/missing'
gmake[1]: Leaving directory '/tmp/postgresql-13.3/config'
PostgreSQL installation complete.
root@solaris:/tmp/postgresql-13.3#
```

We added PostgreSQL to the PATH

```
root@solaris:/tmp/postgresql-13.3# echo 'export PATH=/usr/local/pgsql/bin:$PATH' >> ~/.bashrc
root@solaris:/tmp/postgresql-13.3# source ~/.bashrc
root@solaris:/tmp/postgresql-13.3#
```

Verify that PostgreSQL was installed successfully

```
root@solaris:/tmp/postgresql-13.3# psql --version
psql (PostgreSQL) 13.3
root@solaris:/tmp/postgresql-13.3#
```

```
GNU nano 2.9.3 /etc/apache2/2.4/httpd.conf Modified

<IfDefine !prefork>
    #LoadModule cgid_module libexec/mod_cgid.so
</IfDefine>
</IfModule>
<IfModule mpm_prefork_module>
<IfDefine prefork>
    #LoadModule cgi_module libexec/mod_cgi.so
</IfDefine>
</IfModule>
#LoadModule dav_fs_module libexec/mod_dav_fs.so
#LoadModule dav_lock_module libexec/mod_dav_lock.so
#LoadModule vhost_alias_module libexec/mod_vhost_alias.so
#LoadModule negotiation_module libexec/mod_negotiation.so
LoadModule dir_module libexec/mod_dir.so
#LoadModule actions_module libexec/mod_actions.so
#LoadModule speling_module libexec/mod_speling.so
#LoadModule userdir_module libexec/mod_userdir.so
LoadModule alias_module libexec/mod_alias.so
#LoadModule rewrite_module libexec/mod_rewrite.so

# --- Habilitar soporte para PHP 7 ---
LoadModule php7_module /usr/apache2/2.4/libexec/libphp7.so
AddType application/x-httpd-php .php
DirectoryIndex index.php index.html

<IfModule unixd_module>
```

Then restart apache and create a file example

```
root@solaris:/# svcadm restart apache24
root@solaris:/# |
root@solaris:/# echo "<?php phpinfo(); ?>" > /var/apache2/2.4/htdocs/info.php
root@solaris:/#
```

Now we enable the apache service

```
root@solaris:/tmp# svcadm enable svc:/network/http:apache24
root@solaris:/tmp# svcs -a | grep apache
disabled      11:22:33      svc:/system/apache-stats-24:default
online        12:52:08      svc:/network/http:apache24
root@solaris:/tmp#
```

Go to the browser and search this file created

← → ↻ ⚠ Not secure 10.2.77.35/info.php ☆ ⓘ Relaunch to update

PHP Version 7.1.17



System	SunOS solaris 5.11 11.4.0.15.0 i86pc
Build Date	Aug 10 2018 05:09:27
Configure Command	'/builds/ulhg/workspace/Solaris_11u4/Userland/full-build/02a-build-i386/components/php/php71/php-7.1.17/configure' '--prefix=/usr/php/7.1' '--mandir=/usr/php/7.1/share/man' '--bindir=/usr/php/7.1/bin' '--sbindir=/usr/php/7.1/sbin' '--libdir=/usr/php/7.1/lib' '--enable-fpm' '--with-apxs2=/usr/apache2/2.4/bin/apxs' '--with-zlib' '--with-mysql=shared,mysqlnd' '--with-pdo-mysql=shared,mysqlnd' '--with-mysql-sock=/tmp/mysql.sock' '--enable-bcmath=shared' '--with-bz2=shared' '--enable-calendar=shared' '--with-curl=shared' '--enable-dba=shared' '--with-db4' '--with-cdb' '--with-ldap=shared' '--enable-xml' '--enable-xmlrpc' '--enable-ftp=shared' '--with-gettext=shared' '--with-gd=shared' '--enable-gd-jis-conv' '--enable-gd-native-ttf' '--with-freetype-dir=/usr' '--with-xpm-dir=/usr' '--with-jpeg-dir=/usr' '--with-gmp=shared' '--enable-intl=shared' '--with-ldap=shared' '--enable-mbstring=shared' '--with-openssl=shared' '--with-kerberos' '--enable-pcntl=shared' '--with-readline=shared' '--enable-shmop=shared' '--with-snmp=shared' '--enable-soap=shared' '--enable-sockets=shared' '--enable-sysvsem=shared' '--enable-sysvshm=shared' '--enable-sysvmsg=shared' '--with-tidy=shared' '--enable-wddx=shared' '--with-xmlrpc=shared' '--with-xsl=shared' '--enable-zip=shared' '--with-mhash' '--with-pcre-dir=/usr' '--enable-dtrace' '--disable-rpath' '--sysconfdir=/etc/php/7.1' '--localstatedir=/var/php/7.1' '--with-config-file-path=/etc/php/7.1' '--with-config-file-scan-dir=/etc/php/7.1/conf.d' '--with-pear=/var/php/7.1/pear' '--disable-phpdbg' '--disable-phpdbg-webhelper' '--disable-phpdbg-debug' 'CXXFLAGS=-m64 -O3 -std=c++11' 'CFLAGS=-m64 -O3 -DOPENSSL_NO_SSL2 -DOPENSSL_NO_SSL3'
Server API	Apache 2.0 Handler
Virtual Directory Support	enabled
Configuration File (php.ini) Path	/etc/php/7.1
Loaded Configuration File	/etc/php/7.1/php.ini
Scan this dir for additional .ini files	/etc/php/7.1/conf.d
Additional .ini files parsed	/etc/php/7.1/conf.d/extensions.ini
PHP API	20160303
PHP Extension	20160303
Zend Extension	320160303
Zend Extension Build	API320160303,TS
PHP Extension Build	API20160303,TS
Debug Build	no
Thread Safety	enabled
Zend Signal Handling	enabled
Zend Memory Manager	enabled
Zend Multibyte Support	disabled
IPv6 Support	enabled
DTrace Support	available, disabled
Registered PHP Streams	compress.zlib, php, file, glob, data, http, ftp, phar
Registered Stream Socket Transports	tcp, udp, unix, udg
Registered Stream Filters	zlib.*, convert.iconv.*, string.rot13, string.toupper, string.tolower, string.strip_tags, convert.*, consumed, dechunk

Now we enable the PHP ↔ PostgreSQL connection. Since we don't have any pre-compiled packages, we'll add it from the PHP source code. Download the source code for the same version of PHP you have installed.

```
root@solaris:/tmp# wget https://www.php.net/distributions/php-7.1.33.tar.gz
--2025-11-07 13:16:53-- https://www.php.net/distributions/php-7.1.33.tar.gz
Resolving www.php.net (www.php.net)... 185.85.0.29, 2a02:cb40:200::1ad
Connecting to www.php.net (www.php.net)|185.85.0.29|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 19716547 (19M) [application/octet-stream]
Saving to: 'php-7.1.33.tar.gz'

php-7.1.33.tar.gz      100%[=====] 18.80M  1.44MB/s   in 13s

2025-11-07 13:17:08 (1.41 MB/s) - 'php-7.1.33.tar.gz' saved [19716547/19716547]

root@solaris:/tmp# tar -xzf php-7.1.33.tar.gz
root@solaris:/tmp# cd php-7.1.33/ext/pgsql
root@solaris:/tmp/php-7.1.33/ext/pgsql# phpize
```

Now we install a system build tool that phpize needs: autoconf.

```
root@solaris:/tmp/php-7.4.0/ext/pgsql# pkg install developer/build/autoconf
   Packages to install: 2
   Services to change: 1
   Create boot environment: No
   Create backup boot environment: No

DOWNLOAD                                PKGS      FILES    XFER (MB)   SPEED
Completed                              2/2       75/75     1.2/1.2    456k/s

PHASE                                ITEMS
Installing new actions                 119/119
Updating package state database         Done
Updating package cache                  0/0
Updating image state                    Done
Creating fast lookup database           Done
Updating package cache                  1/1
root@solaris:/tmp/php-7.4.0/ext/pgsql#
```

We repeat the compilation process. Next do make and make install

```
root@solaris:/tmp/php-7.1.33/ext/pgsql# phpize
Configuring for:
PHP Api Version:           20160303
Zend Module Api No:        20160303
Zend Extension Api No:     320160303
root@solaris:/tmp/php-7.1.33/ext/pgsql# ./configure --with-pgsql=/usr/local/pgsql
checking for grep that handles long lines and -e... /usr/bin/ggrep
checking for egrep... /usr/bin/ggrep -E
checking for a sed that does not truncate output... /usr/bin/gsed
checking for cc... no
checking for gcc... gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... no
checking for suffix of object files... o
checking whether we are using the GNU C compiler... yes
checking whether gcc accepts -g... yes
checking for gcc option to accept ISO C89... none needed
checking how to run the C preprocessor... gcc -E
checking for icc... no
checking for suncc... no
checking whether gcc and cc understand -c and -o together... yes
checking for system library directory... lib
checking if compiler supports -R... yes
checking build system type... x86_64-pc-solaris2.11
```

```

root@solaris:/tmp/php-7.1.33/ext/pgsql# make
/bin/sh /tmp/php-7.1.33/ext/pgsql/libtool --mode=compile gcc -DZEND_ENABLE_STATIC_TSRMLS_CACHE=1 -I. -I/tmp/php-7.1.33/ext/pgsql -DPHP_ATOM_INC -I/tmp/php-7.1.33/ext/pgsql/include -I/tmp/php-7.1.33/ext/pg
sql/main -I/tmp/php-7.1.33/ext/pgsql -I/usr/php/7.1/include/php -I/usr/php/7.1/include/php/main -I/usr
/php/7.1/include/php/TSRM -I/usr/php/7.1/include/php/Zend -I/usr/php/7.1/include/php/ext -I/usr/php/7.
1/include/php/ext/date/lib -I/usr/local/pgsql/include -DHAVE_CONFIG_H -g -O2 -c /tmp/php-7.1.33/ex
t/pgsql/pgsql.c -o pgsql.lo
mkdir .libs
gcc -DZEND_ENABLE_STATIC_TSRMLS_CACHE=1 -I. -I/tmp/php-7.1.33/ext/pgsql -DPHP_ATOM_INC -I/tmp/php-7.1
.33/ext/pgsql/include -I/tmp/php-7.1.33/ext/pgsql/main -I/tmp/php-7.1.33/ext/pgsql -I/usr/php/7.1/incl
ude/php -I/usr/php/7.1/include/php/main -I/usr/php/7.1/include/php/TSRM -I/usr/php/7.1/include/php/Zen
d -I/usr/php/7.1/include/php/ext -I/usr/php/7.1/include/php/ext/date/lib -I/usr/local/pgsql/include -D
HAVE_CONFIG_H -g -O2 -c /tmp/php-7.1.33/ext/pgsql/pgsql.c -fPIC -DPIC -o .libs/pgsql.o
In file included from /tmp/php-7.1.33/ext/pgsql/php_pgsql.h:58:0,
      from /tmp/php-7.1.33/ext/pgsql/pgsql.c:49:
/usr/local/pgsql/include/pg_config.h:766:0: warning: "PACKAGE_URL" redefined
#define PACKAGE_URL "https://www.postgresql.org/"

In file included from /usr/php/7.1/include/php/TSRM/tsrm_config.h:1:0,
      from /usr/php/7.1/include/php/TSRM/tsrm_config_common.h:13,
      from /usr/php/7.1/include/php/Zend/zend_virtual_cwd.h:27,
      from /usr/php/7.1/include/php/main/php.h:411,
      from /tmp/php-7.1.33/ext/pgsql/pgsql.c:35:
/usr/php/7.1/include/php/main/./main/php_config.h:2190:0: note: this is the location of the previous
definition
#define PACKAGE_URL ""

In file included from /tmp/php-7.1.33/ext/pgsql/php_pgsql.h:58:0,
      from /tmp/php-7.1.33/ext/pgsql/pgsql.c:49:
/usr/local/pgsql/include/pg_config.h:969:0: warning: "restrict" redefined
#define restrict __restrict

In file included from /usr/php/7.1/include/php/Zend/zend_types.h:27:0,

root@solaris:/tmp/php-7.1.33/ext/pgsql# make install
Installing shared extensions:      /usr/php/7.1/lib/extensions/no-debug-zts-20160303/
root@solaris:/tmp/php-7.1.33/ext/pgsql#

```

On the info.php page, we look for the line: Loaded Configuration File and check the exact path

← → ↺

Not secure 10.2.77.35/info.php

Relaunch to update

Loaded Configuration File 1/1

System	SunOS solaris 5.11 11.4.0.15.0 i86pc
Build Date	Aug 10 2018 05:09:27
Configure Command	'/builds/ulhg/workspace/Solaris_11u4/Userland/full-build/02a-build-i386/components/php/php71/php-7.1.17/configure' '--prefix=/usr/php/7.1' '--mandir=/usr/php/7.1/share/man' '--bindir=/usr/php/7.1/bin' '--sbindir=/usr/php/7.1/sbin' '--libdir=/usr/php/7.1/lib' '--enable-fpm' '--with-apxs2=/usr/apache2/2.4/bin/apxs' '--with-zlib' '--with-mysql=shared,mysqlnd' '--with-pdo-mysql=shared,mysqlnd' '--with-mysql-sock=/tmp/mysql.sock' '--enable-bcmath=shared' '--with-bz2=shared' '--enable-calendar=shared' '--with-curl=shared' '--enable-dba=shared' '--with-db4' '--with-cdb' '--with-ldap=shared' '--enable-xmlrpc=shared' '--with-openssl=shared' '--with-gd=shared' '--enable-gd-jis-conv' '--enable-gd-native-ttf' '--with-freetype-dir=/usr' '--with-xpm-dir=/usr' '--with-jpeg-dir=/usr' '--with-gmp=shared' '--enable-intl=shared' '--with-ldap=shared' '--enable-mbstring=shared' '--with-openssl=shared' '--with-kerberos' '--enable-pcntl=shared' '--with-readline=shared' '--enable-shmop=shared' '--with-snmp=shared' '--enable-soap=shared' '--enable-sockets=shared' '--enable-sysvsem=shared' '--enable-sysvshm=shared' '--enable-sysvmsg=shared' '--with-tidy=shared' '--enable-wddx=shared' '--with-xmlrpc=shared' '--with-xsl=shared' '--enable-zip=shared' '--with-mhash' '--with-pcre-dir=/usr' '--enable-dtrace' '--disable-rpath' '--sysconfdir=/etc/php/7.1' '--localstatedir=/var/php/7.1' '--with-config-file-path=/etc/php/7.1' '--with-config-file-scan-dir=/etc/php/7.1/conf.d' '--with-pear=/var/php/7.1/pear' '--disable-phpdbg' '--disable-phpdbg-webhelper' '--disable-phpdbg-debug' 'CXXFLAGS=-m64 -O3 -std=c++11' 'CFLAGS=-m64 -O3 -DOPENSSL_NO_SSL2 -DOPENSSL_NO_SSL3'
Server API	Apache 2.0 Handler
Virtual Directory Support	enabled
Configuration File (php.ini) Path	/etc/php/7.1
Loaded Configuration File	/etc/php/7.1/php.ini
Scan this dir for additional .ini files	/etc/php/7.1/conf.d
Additional .ini files parsed	/etc/php/7.1/conf.d/extensions.ini

We added the pgsql extension to that php.ini file. and restart apache

```
Command Prompt - ssh santi x + v
GNU nano 2.9.3 /etc/php/7.1/php.ini Modified
;extension=php_pdo_odbc.dll
;extension=php_pdo_pgsql.dll
;extension=php_pdo_sqlite.dll
;extension=php_pgsql.dll
;extension=php_shmop.dll
extension=/usr/php/7.1/lib/extensions/no-debug-zts-20160303/pgsql.so
; The MIBS data available in the PHP distribution must be installed.
; See http://www.php.net/manual/en/snmp.installation.php
;extension=php_snmp.dll

root@solaris:/tmp/php-7.1.33/ext/pgsql# svcadm restart apache24
root@solaris:/tmp/php-7.1.33/ext/pgsql#
```

Then we verified that the extension was loaded on the page.

pgsql

PostgreSQL Support	enabled
PostgreSQL(libpq) Version	13.3
PostgreSQL(libpq)	PostgreSQL 13.3 on x86_64-pc-solaris2.11, compiled by gcc (GCC) 7.3.0, 64-bit
Multibyte character support	enabled
SSL support	disabled
Active Persistent Links	0
Active Links	0

b. Create a simple application that does the following:

- Connect to the PostgreSQL database created in the previous lab.
- Calculate the students' grade point average.
- Store student results in the database.
- It must allow consultation of a student's grade records based on their ID number.

First we made sure that Solaris could reach PostgreSQL on Slackware using ping, and then we tested port 5432 (PostgreSQL):

```
root@solaris:/tmp/php-7.1.33/ext/pgsql# ping 10.2.77.31
10.2.77.31 is alive
root@solaris:/tmp/php-7.1.33/ext/pgsql# telnet 10.2.77.31 5432
Trying 10.2.77.31...
Connected to 10.2.77.31.
Escape character is '^]'.
^]
telnet> |
```

On Linux Slackware, enter on postgres

```
root@aysrSlack:~# su - postgres

God grant us the serenity to accept the things we cannot change, courage to
change the things we can, and wisdom to know the difference.

postgres@aysrSlack:~$ psql
Password for user postgres:
psql: error: connection to server on socket "/tmp/.s.PGSQL.5432" failed: FATAL: password authentica
tion failed for user "postgres"
postgres@aysrSlack:~$ psql
Password for user postgres:
psql (14.19)
Type "help" for help.

postgres=#
```

Now we create the database and the table within this database

```
postgres=# CREATE DATABASE notas;
CREATE DATABASE
postgres=# \c notas;
You are now connected to database "notas" as user "postgres".
notas=# CREATE TABLE estudiantes (
notas=# ^C
notas=# CREATE TABLE estudiantes (id VARCHAR(20) PRIMARY KEY, name VARCHAR(100), note1 NUMERIC(4,2),
note2 NUMERIC(4,2), note3 NUMERIC(4,2), average NUMERIC(4,2));
CREATE TABLE
notas=# _
```

Now in Solaris, inside the web server folder:

```
root@solaris:/# cd /var/apache2/2.4/htdocs/
root@solaris:/var/apache2/2.4/htdocs# |
```

Create a new page in index.php

```
GNU nano 2.9.3 index.php Modified
<?php
// Configuración de conexión
$host = "10.2.77.31"; // IP de tu servidor Slackware
$port = "5432";
$dbname = "notas";
$user = "postgres";
$password = "admin123";

$conn = pg_connect("host=$host port=$port dbname=$dbname user=$user password=$password");

if (!$conn) {
    die("<p style='color:red'> Error al conectar a la base de datos.</p>");
}

// Registrar estudiante
if (isset($_POST['guardar'])) {
    $id = $_POST['id'];
    $name = $_POST['name'];
    $note1 = $_POST['note1'];
    $note2 = $_POST['note2'];
    $note3 = $_POST['note3'];

    // Calcular promedio
    $average = ($note1 + $note2 + $note3) / 3;

    // Guardar en la BD
    $result = pg_query_params($conn,
        "INSERT INTO estudiantes (id, name, note1, note2, note3, average)
        VALUES ($1,$2,$3,$4,$5,$6)
        ON CONFLICT (id) DO UPDATE SET
        name=$2, note1=$3, note2=$4, note3=$5, average=$6",
        array($id, $name, $note1, $note2, $note3, $average)
    );

    if ($result) {
        echo "<p style='color:green'> Estudiante guardado con promedio $average</p>";
    } else {
        echo "<p style='color:red'> Error al guardar estudiante.</p>";
    }
}

// Consultar estudiante
if (isset($_GET['buscar'])) {
    $id = $_GET['id'];
    $result = pg_query_params($conn, "SELECT * FROM estudiantes WHERE id = $1", array($id));
```

Go to index.php

← → ↻ ⚠ Not secure 10.2.77.35/index.php ☆ ⓘ Relaunch to update ⋮

Registrar Estudiante

ID:
Nombre:
Nota 1:
Nota 2:
Nota 3:

Consultar Estudiante

ID:

We tested: Registering a student (it will show you the average). Searching for a student by ID.

← → ↻ ⚠ Not secure 10.2.77.35/index.php?id=123&buscar=Buscar 🖨 ☆ ⓘ Relaunch to update ⋮

Resultado de la búsqueda:

ID: 123
Nombre: Santiago
Notas: 50.00, 50.00, 40.00
Promedio: **46.67**

Registrar Estudiante

ID:
Nombre:
Nota 1:
Nota 2:
Nota 3:

Consultar Estudiante

ID:

3. Hosting Service Configuration

On the machine with the Solaris operating system, configure the virtual host service so that hosting services can be provided to multiple companies. The service should be configured in such a way that three webpages hosted on the same web server can be accessed by name, using the URLs of two different domains.

Create virtual hosts that can be accessed as shown below, all of which will reach the same web server installed on Solaris, but each will direct to different webpages.

- network..com.co
- security..org.jp

Show the results to your professor.

Hint: To perform this configuration, you must

- a. Configure the virtual host service in Apache.
- b. Configure the DNS service so that the server names point to the same IP address. Keep in mind that two of the entries belong to one domain and the other to a different domain, so you will need to configure this from the primary DNS servers of each domain.

In Solaris, create one folder for each domain.

```
root@solaris:/# cd /var/apache2/2.4/htdocs
root@solaris:/var/apache2/2.4/htdocs# mkdir -p network security
root@solaris:/var/apache2/2.4/htdocs# |
```

We created an index.html file in each folder.

```
root@solaris:/var/apache2/2.4/htdocs# echo "<h1>Network site - Santiago</h1>" > /var/apache2/2.4/htdocs/network/index.html
root@solaris:/var/apache2/2.4/htdocs# echo "<h1>Security site - Julian</h1>" > /var/apache2/2.4/htdocs/security/index.html
root@solaris:/var/apache2/2.4/htdocs# |
```

Now, we created the file: httpd-vhosts.conf

```
GNU nano 2.9.3 /etc/apache2/2.4/conf/extra/httpd-vhosts.conf Modified
# Virtual Host 1 - network.santiago.com.co
<VirtualHost *:80>
    ServerName network.santiago.com.co
    DocumentRoot "/var/apache2/2.4/htdocs/network"
    ErrorLog "logs/network-error.log"
    CustomLog "logs/network-access.log" common
</VirtualHost>

# Virtual Host 2 - security.julian.org.jp
<VirtualHost *:80>
    ServerName security.julian.org.jp
    DocumentRoot "/var/apache2/2.4/htdocs/security"
    ErrorLog "logs/security-error.log"
    CustomLog "logs/security-access.log" common
</VirtualHost>
```

Conclusions

Throughout this laboratory session, we explored the essential components of network and server infrastructure used in modern organizations. First, we configured routers in a simulated environment using Cisco Packet Tracer, applying fundamental IOS commands, addressing schemes, static routing, console access configurations, and device security practices. These exercises strengthened our understanding of Layer 2 and Layer 3 communication, routing behavior, end-to-end connectivity, and troubleshooting techniques using tools such as *ping* and *traceroute*.

In the second part of the lab, we installed and configured web services across different operating systems, including Solaris, Slackware Linux, and Windows Server. We deployed Apache and Nginx, validated their functionality through simple test pages, enabled automatic

service startup, and implemented DNS resolution to provide name-based access to each server. For multi-tenant environments, we configured virtual hosts to serve multiple websites from the same server, demonstrating how hosting providers isolate and deliver content based on domain names.

Finally, we integrated web applications with a PostgreSQL database by enabling PHP extensions, allowing dynamic content generation, grade calculations, and student data storage. This reinforced the interaction between the application layer, database services, and the underlying operating system.

Overall, the laboratory provided hands-on experience with real-world networking, server administration, name resolution, and web hosting concepts. By combining routing, DNS, web services, and application development, we gained a comprehensive understanding of how core infrastructure components interoperate to support organizational IT services.

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